GATE

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EC : ELECTRONICS AND COMMUNICATIONS
EE : ELECTRICAL ENGINEERING

SIGNALS AND SYSTEMS

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Topic 1 : Linear Time Invariant System

PART(A) : CONTINUOUS TIME SIGNALS AND SYSTEM

Introduction

Signals : A signal is a single valued function of time that represents a physical variable of interest such as voltage, current, displacement, velocity etc. It can be either real valued or complex valued.

System : A system is a combination and interconnection of several components to perform a desired task. Systems may be interconnected with each other to form other systems. In such cases the component systems are called as *subsystems*.

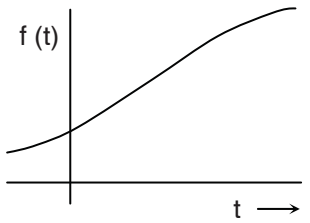
CLASSIFICATION OF SIGNALS

There are several classes of signals.

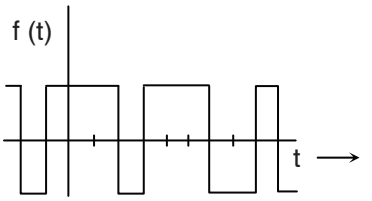
1. Continuous-time and discrete-time signals
2. Analog and digital signals
3. Periodic and aperiodic signals
4. Energy and power signals
5. Deterministic and probabilistic signals

1. *Continuous-time and discrete-time signals* **:**

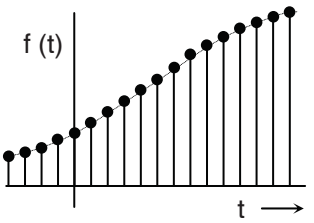
   A signal that is specified for every value of time t is a *continuous time signal*, (fig. (a) and (b)) and a signal that is specified only at discrete values of time t is a *discrete time signal* (fig. (c) and (d)).



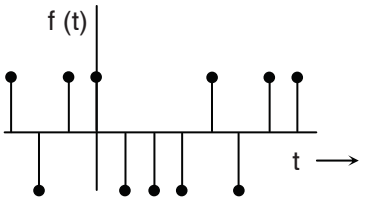
(a) Analog, continuous time



(b) Digital, continuous time



(c) Analog, discrete time



(d) Digital, discrete time

Examples of Signals

2. *Analog and digital signals* :

Analog signal :

The concept of continuous time is often confused with that of analog. The two are not the same. The same is true of the concepts of discrete time and digital. A signal whose amplitude can take on any value in a continuous range is an analog signal (fig. (a) and (c)). A *digital signal*, on the other hand, is one whose amplitude can take on only a finite number of values (fig. (b) and (d)). Signals associated with a digital computer are digital because they take on only two values (binary signals).

Thus the terms continuous time and discrete time qualify the nature of a signal along the time (horizontal) axis. The terms analog and digital, on the other hand, qualify the nature of the signal amplitude (vertical axis). An analog signal can be converted into a digital signal [analog–to–digital (A/D) conversion] through quantization (rounding off).

3. *Periodic and aperiodic (non-periodic)signals* :

A signal f(t) is said to be periodic if for some positive constant T_0

$$f(t) = f(t + T_0) \qquad \text{for all } t \qquad \ldots(1)$$

The smallest value of T_0 that satisfies the periodicity condition (1) is the period of f(t). A signal for which there is no value of T_0 for which equation (1) is satisfied are called aperiodic (non-periodic) signal.

Note: A periodic signal, by definition, is an everlasting signal.

A signal that does not start before t = 0 is a *causal* signal.

i.e. $f(t) = 0, \quad t < 0$

A signal that starts before t = 0 is a *noncausel* signal. A signal that is zero for all $t \geq 0$ is called an *anticausal* signal.

Note: Observe that an everlasting signal is always noncausal but a noncausal signal is not necessarily everlasting.

Example

Determine if the signal is periodic, if yes then find its fundamental frequency and period

1. $f_1(t) = 2 + 7\cos\left(\frac{1}{2}t + \theta_1\right) + 3\cos\left(\frac{2}{3}t + \theta_2\right) + 5\cos\left(\frac{7}{6}t + \theta_3\right)$

2. $f_2(t) = 2\cos(2t + \theta_1) + 5\sin(\pi t + \theta_2)$

Solution :

1. $f_1(t) = 2 + 7\cos\left(\frac{1}{2}t+\theta_1\right)+3\cos\left(\frac{2}{3}t+\theta_2\right)+5\cos\left(\frac{7}{6}t+\theta_3\right)$

   In order for the signal to be periodic the ratios of their frequencies must be a rational number here

   $$\omega_1 = \frac{1}{2}, \qquad \omega_2 = \frac{2}{3} \qquad \text{and} \qquad \omega_3 = \frac{7}{6}$$

   $\omega_{fundamental} = \text{GCD}\ (\omega_1, \omega_2, \omega_3)$ …(GCD ≡ greatest common divisor)

   $$= \frac{1}{6}(\text{rational}) \Rightarrow f_{fundamental} = \frac{\omega}{2\pi} = \frac{1}{12\pi}$$

   Thus, given signal is periodic with fundamental frequency $\frac{1}{6}$ rad/sec $\left(\frac{1}{12\pi}\text{Hz}\right)$ and period (12π seconds)

2. $f_2(t) = 2\cos(2t+\theta_1)+5\sin(\pi t+\theta_2)$

   The signal is not periodic

   $\because$ ratio of its frequency i.e. $\frac{\omega_1}{\omega_2} = \frac{2}{\pi}$ is irrational

4. *Energy and power signals* :

   (i) Energy Signals : The signal energy is given by

   $$E = \int_{-\infty}^{\infty} f^2(t).\,dt \qquad \ldots(2)$$

   in case of complex signal

   $$E = \int_{-\infty}^{\infty} |f(t)|^2\, dt \qquad \ldots(3)$$

   Note: For signal to be energy signal E must be finite i.e. $E < \infty$. Also as $t \to \infty$, $f(t) \to 0$.

   Exemples

   1. $f(t) = e^{-at}\ u(t)$

      $$E = \int_{-\infty}^{\infty} e^{-2at}.u(t)dt = \int_{0}^{\infty} e^{-2at}dt = \left[\frac{e^{-2at}}{-2a}\right]_0^{\infty} = \frac{1}{2a} \rightarrow \text{Energy signal}$$

   2. $f(t) = \delta(t)$

      $$E = \int_{-\infty}^{\infty} \delta^2(t)\ dt = 1 \rightarrow \text{Energy signal}$$

   3. $f(t) = A, \quad -\frac{\tau}{2} \le t \le \frac{\tau}{2}$

      $$E = \int_{-\tau/2}^{\tau/2} A^2\ dt = \left[A^2(t)\right]_{-\tau/2}^{\tau/2} = A^2\tau \rightarrow \text{Energy signal}$$

(ii) *Power Signal* :

The signal power is given by

$$P = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} f^2(t)dt$$

We can generalize this definition for a complex signal f(t) as

$$P = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} |f(t)|^2 \, dt$$

Observe that the signal power P is the time average (mean) of the signal amplitude squared, that is, the mean-squared value of f(t). Indeed, the square root of P is the familiar **rms** (root mean square) value of f(t).

- Energy signal has zero average power, while power signal has infinite energy.
- A signal cannot both be energy and a power signal. If it is one, it cannot be the other. On the other hand, there are signals that are neither energy nor power signals. The ramp signal is such an example.
- Signals which are either periodic or random are power signals, where as signals which are both aperiodic and deterministic are energy signals.

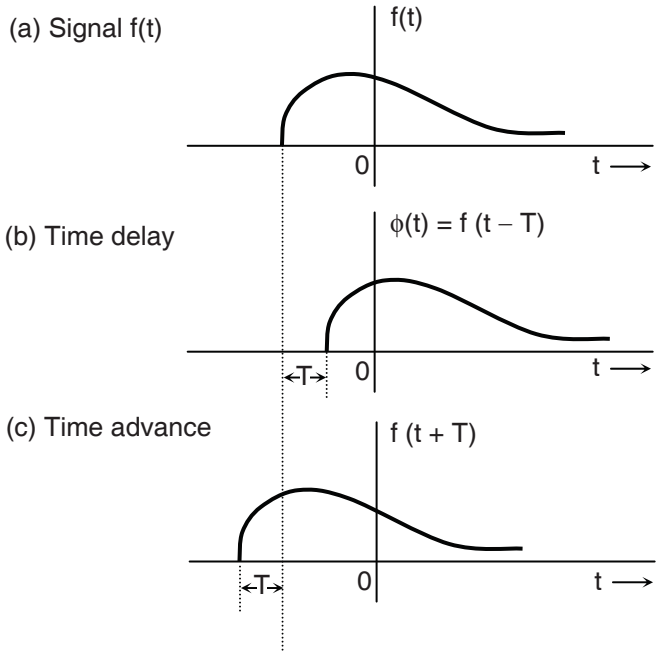
5. *Deterministic and probabilistic signals* :

A signal whose physical description is known completely, either in a mathematical form or a graphical form, is a *deterministic signal*. A signal whose values cannot be predicted precisely but are known only in terms of probabilistic description, such as mean value, mean squared value, and so on is a *random signal*.

USEFUL SIGNAL OPERATIONS

(A) Time Shifting

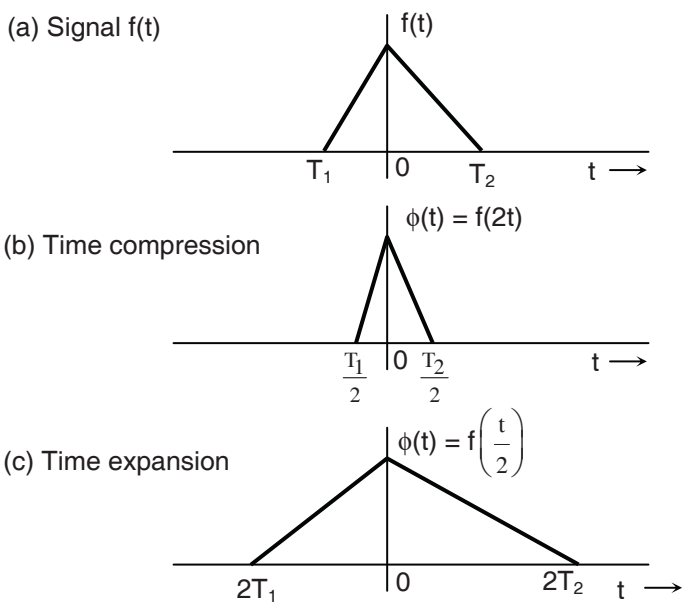
Consider a signal f(t) shown in figure (a) below. To time shift the signal by T seconds, we replace t with t –T which is represented by $\phi(t) = f(t - T)$. If T is positive, the shift is to the right (delay) (fig. (b)). If T is negative, the shift is to the left (advance) (fig. (c)).



Time shifting a signal

(B) Time Scaling

The compression or expansion of a signal in time is known as *time scaling*. To time scale a signal by a factor *a*, we replace *t* with *at*. If $a > 1$, the scaling results in *compression*, and if $a < 1$, the scaling results in *expansion*. For example if f(t) were recorded on a tape and played back at twice the normal recording speed, we would obtain f (2t) i.e. f(t) compressed by factor of 2.

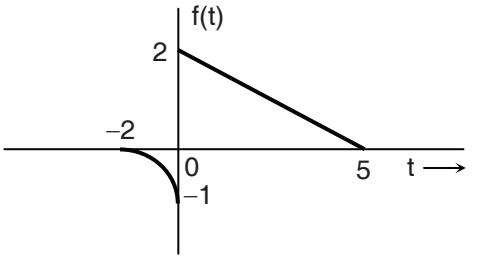


Time Scaling of a Signal

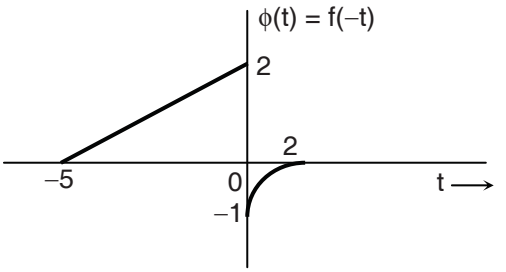
Note : Observe that in time scaling operation, the origin t = 0 is the anchor point, which remains unchanged under scaling operation because at t = 0, f(t) = f (at) = f (0).

(C) Time Inversion (Time Reversal)

To time-invert a signal we replace *t* with –*t*, which can be obtained by taking mirror image of f(t) about vertical axis.



(a) Signal f(t)



(b) Time inversion

Time inversion (reflection) of a signal

Note : The mirror image of f(t) about the vertical axis is f(–t). Recall also that the mirror image of f(t) about the horizontal axis is –f(t).

Combined Operations

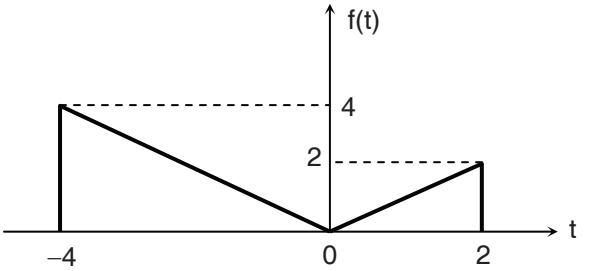
Certain complex operations require simultaneous use of more than one of the above operation. The most general operation involving all the three operations is f (at – b), which is realized in two possible sequences of operation :

1. Time-shift f(t) by *b* to obtain f(t – b). Now time-scale the shifted signal f(t – b) by *a* (i.e. replace *t* with *at*) to obtain f (at – b).
2. Time scale f(t) by *a* to obtain f(at). Now time-shift f(at) by $\frac{b}{a}$, (i.e. replace t with $\left(t-\frac{b}{a}\right)$ to obtain $f\left[a\left(t-\frac{b}{a}\right)\right] = f(at-b)$).

In either case, if *a* is negative, time scaling involves time inversion.

Example :

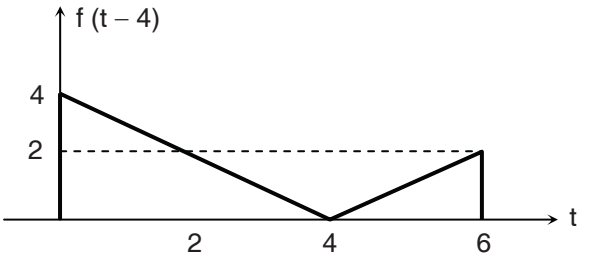
For given signal f(t) find f(2t – 4)



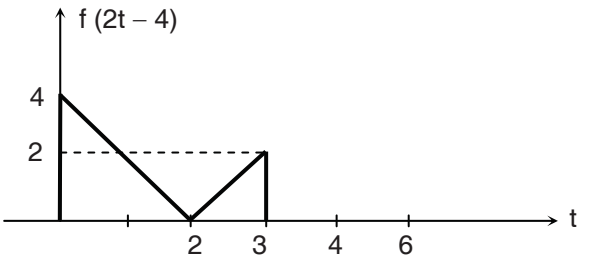
Solution :

Method 1 :

Step 1 : Time shifting f (t – 4) ⇒ f(t) is delayed by 4 seconds. (Shift left)

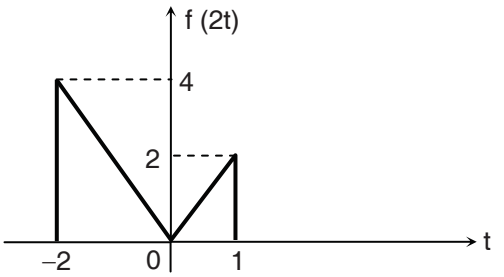


Step 2 : Time scaling f (2t – 4)
⇒ scaling f(t – 4) by factor 2 (compression)

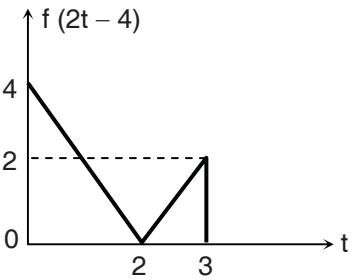


Method 2 :

Step 1 : Time scaling f (2t)
⇒ compresession by factor 2



Step 2 : Time shifting f [2 (t–2)] = f(2t – 4)
⇒ delaying signal f(2t) by 2 seconds



we get the same result using any of the method

TYPE OF SIGNALS

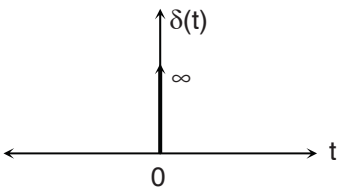
(1) Unit Impulse Function :

Mathematically : An impulse is a unit pulse of extremely large magnitude and infinitesimal duration. As $t \to 0$, magnitude $\to \infty$, such that area is unity. It is defined as

$$\delta(t) = 0 \quad , \quad t \neq 0$$

$$\int_{-\infty}^{\infty} \delta(t)\, dt = 1$$

Representation :



Important Properties of unit function :

- $\delta(at) = \dfrac{1}{|a|}\delta(t)$

- $\delta(-t) = \delta(t)$

- $\left.\begin{aligned} f(t)\,\delta(t) &= x(0)\,\delta(t) \\ f(t)\,\delta(t-t_0) &= x(t_0)\,\delta(t-t_0) \end{aligned}\right\}$ …multiplication property

- $\left.\begin{aligned} &\int_{-\infty}^{\infty} x(t)\,\delta(t)\,dt = x(0) \\ &\int_{-\infty}^{\infty} x(t)\,\delta(t-a) = x(a) \end{aligned}\right\}$ … Sampling property

Note: Integral exists, provided f(t) is continuous at the place where impulse occurs.

Example :

Solve

i) $\left[\sin\left(t^2 - \dfrac{\pi}{2}\right)\right]\delta(t)$

ii) $\displaystyle\int_{-\infty}^{\infty} \delta(t-2)\cos\left(\frac{\pi t}{4}\right)dt$

Solution:

i) $\sin\left(t^2 - \frac{\pi}{2}\right).\delta(t) = \sin\left(-\frac{\pi}{2}\right).\delta(t)$ …[multiplication property of $\delta(t)$]

$= -\delta(t)$

ii) $\int_{-\infty}^{\infty} \delta(t-2).\cos\left(\frac{\pi t}{4}\right)dt$

now $\delta(t-2)$ is impulse delayed by 2 seconds

$\therefore$ by sampling propertry of $\delta(t)$

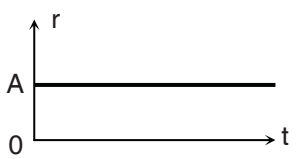
$\int_{-\infty}^{\infty} \delta(t-2).\cos\left(\frac{\pi t}{4}\right)dt = \cos\left(\frac{2\pi}{4}\right)$ …[putting t = 2] = 0

(2) Step input :

Mathematically : $r = 0, \quad t < 0$

$r = A, \quad t \geq 0$

Representation:

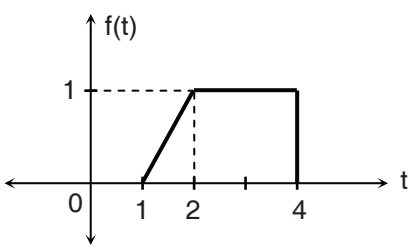


When A = 1, it is called unit step function, denoted by u(t).

The unit step function, u(t) also proves very useful in specifying a function which has different mathematical descriptions over different intervals of times, since such a description often proves clumsy and inconvenient in mathematical treatment. With the help of *unit step* function, we can describe such functions by a single expression that is valid for all *t*.

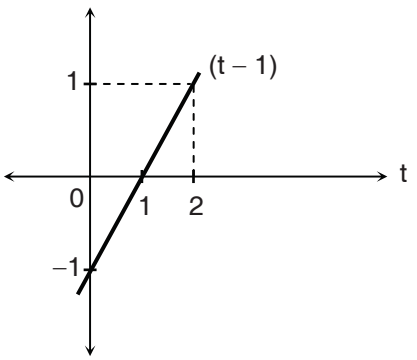
Example :

Obtain a single expression for f(t), which is true for all time 't'

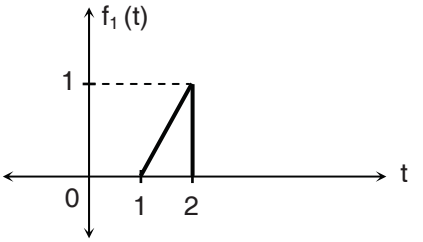


Solution :

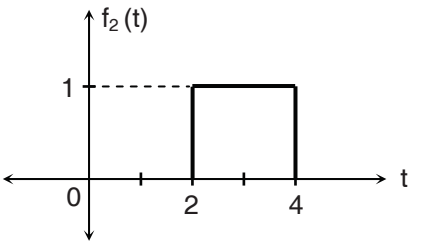
f(t) can be obtain as a sum of two signals $f_1(t)$ and $f_2(t)$ as shown below



$\Rightarrow$



$f_1(t) = (t–1)\ [u(t–1)–u(t–2)]$



$f_2(t) = u(t–2)–u(t–4)$

$$\begin{aligned}
\therefore\ f(t) &= f_1(t) + f_2(t) \\
&= (t-1)\,[u(t-1) - u(t-2)] + u(t-2) - u(t-4) \\
&= (t-1)\,u(t-1) - (t-1)\,u(t-2) + u(t-2) - u(t-4) \\
&= (t-1)\,u(t-1) - (t-2+1)\,u(t-2) + u(t-2) - u(t-4) \\
&= (t-1)\,u(t-1) - (t-2)\,u(t-2) - u(t-4) \qquad \text{...(for all } t\text{)}
\end{aligned}$$

- $u(t) = \int_{-\infty}^{t} \delta(t)\,dt$
- Derivative of a unit step is unit impulse $\frac{d}{dt}u(t) = \delta(t)$
- Derivative of a unit impulse is doublet.
- Derivative of a unit doublet is triplet.

(3) Unit Gate Pulse :

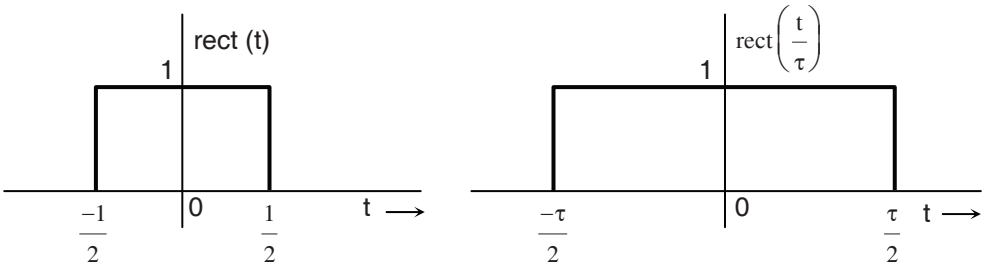
We define a unit gate function rect (t) as a gate pulse of unit height and unit width, centered at the origin, as illustrated in following figure (a).

Mathematically :

$$\text{rect}(t) = \begin{cases} 0 & |t| > \frac{1}{2} \\ \frac{1}{2} & |t| = \frac{1}{2} \\ 1 & |t| < \frac{1}{2} \end{cases}$$

The gate pulse in following figure (b), is the unit gate pulse rect (t) expanded by a factor τ and therefore can be expressed as rect $\left(\frac{t}{\tau}\right)$. Observe that τ, the denominator of the argument of rect $\left(\frac{t}{\tau}\right)$, indicates the width of the pulse.

Representation :



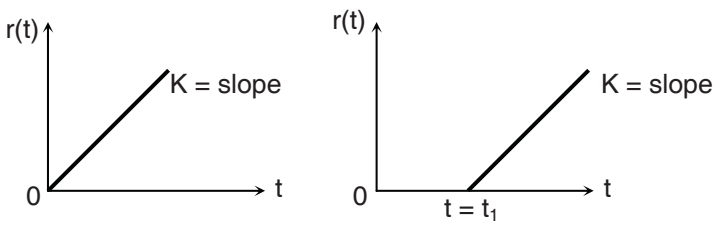
(a) A gate pulse **(b)**

(4) Ramp Function :

Mathematically : $r(t) = 0, \quad t < 0$

$r(t) = Ktu(t), \quad t \geq 0$

Representation :



$r(t) = Kt\,u(t)$ delayed ramp

$r(t) = K(t - t_1)\,u(t - t_1)$

> Derivative of unit ramp(Slope, K = 1) is the unit step signal. i.e. $\frac{dr(t)}{dt} = u(t)$

(5) Unit Triangle Function

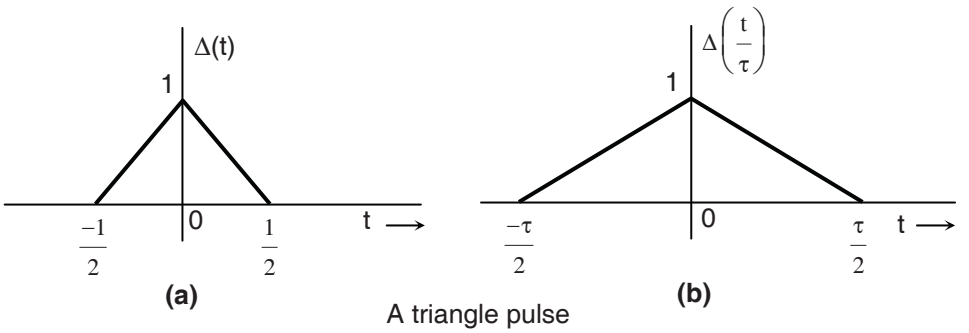
We define a unit triangle function $\Delta(t)$ as a triangular pulse of unit height and unit width, centered at the origin, as shown in following figure.

Mathematically :

$$\Delta(t) = \begin{cases} 0 & |t| \geq \frac{1}{2} \\ 1 - 2|t| & |t| < \frac{1}{2} \end{cases}$$

The pulse in following figure (b) is $\Delta\left(\frac{t}{\tau}\right)$. Here also denominator τ of the argument of $\Delta\left(\frac{t}{\tau}\right)$ indicates the pulse width.

Representation :



A triangle pulse

(6) Sinusoidal input Function :

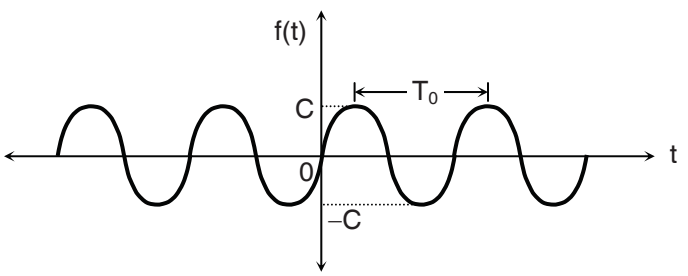
Mathematically : $f(t) = C \sin(2\pi F_0 t + \theta)$

… where C = amplitude

F_0 = Frequency in Hertz

θ = Phase in radians

Representation :



Cleary, this sinusoid repeats every $1/F_0$ seconds. As a result, there are F_0 repetitions per second. This is the frequency of the sinusoid, and the repetition interval T_0 given by

$$T_0 = \frac{1}{F_0} = \frac{2\pi}{\omega_0} \qquad \text{… (Radian frequency, } \omega_0 = 2\pi F_0\text{)}$$

(7) Interpolation Function sinc (t) :

The function sin t/t is the "sine over argument" function denoted by sinc (t). This function plays an important role in signal processing. It is also known as the *filtering or interpolating function*.

Mathematically :

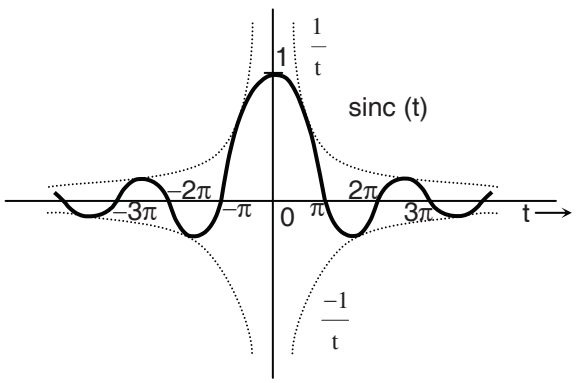
$$\text{sinc}\ (t) = \frac{\sin t}{t} \qquad \text{or} \qquad \text{sinc}\ (t) = \frac{\sin \pi t}{\pi t} \qquad \ldots(4)$$

Inspection of equation (4) shows that

1. sinc (t) is an even function of t.
2. sinc (t) = 0 when sin t = 0 except at t = 0, where it appears indeterminate. This means that sinc t= 0 for t = $\pm\pi$, $\pm 2\pi$, $\pm 3\pi$,…
3. Using L'Hopital's rule, we find sinc (0) = 1.
4. sinc (t) is the product of an oscillating signal sin t (of period 2π) and a monotonically decreasing function 1/t. Therefore, sinc (t) exhibits sinusoidal oscillations of period 2π, with amplitude decreasing continuously as 1/t.

Following figure shows sinc(t). Observe that sinc (t) = 0 for values of t that are positive and negative integral multiples of π.

Representation :



A sinc pulse

(8) Random Function :

The amplitude of this function is random and similar to white noise as it randomly contains all frequencies.

Representation :

(9) The Exponential Function e^{st}

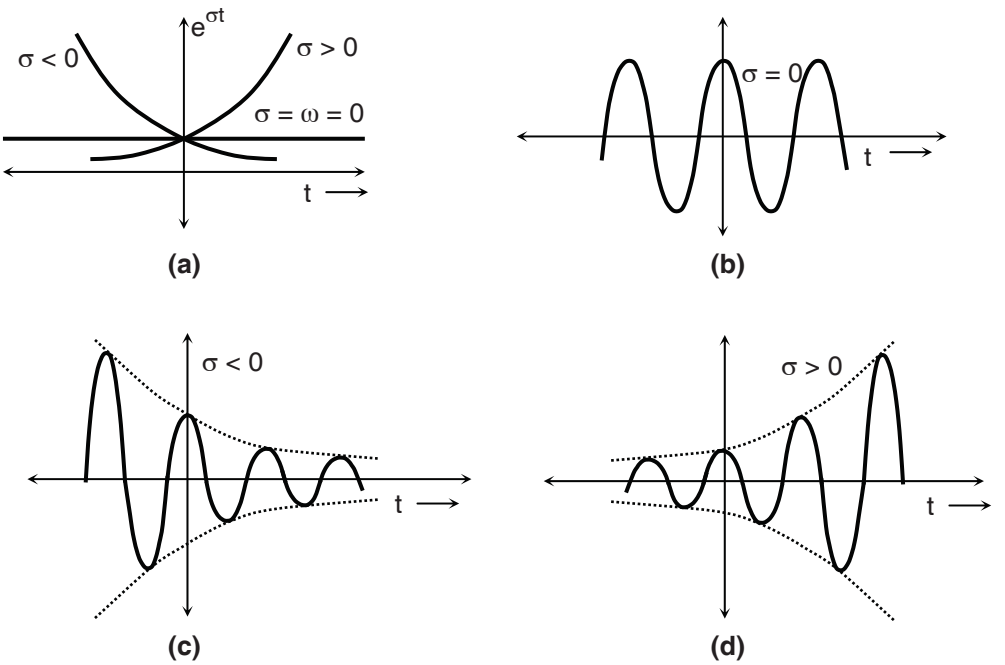
One of the most important functions in the area of signals and systems is the exponential signal e^{st}, where s is complex frequency in general, given by

$$s = \sigma + j\omega$$

Function e^{st} encompasses a large class of functions. The following functions are special cases of e^{st}:

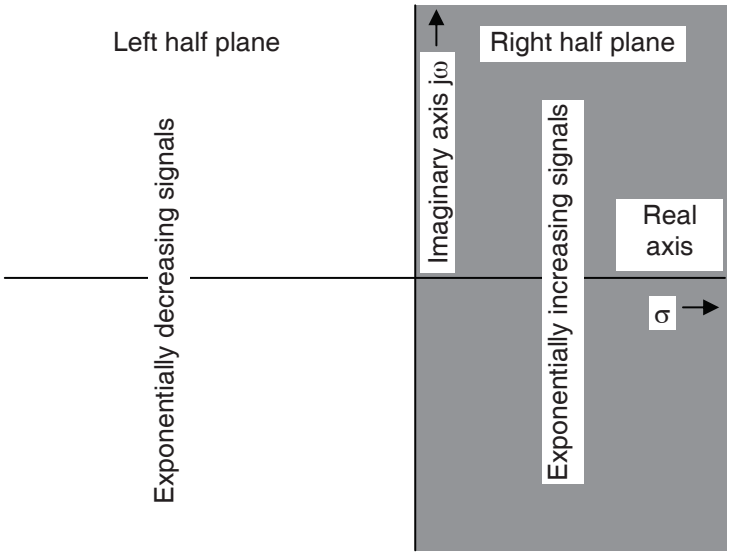
1. A constant $k = ke^{0t}$ $(s = 0)$ …(fig. (a))
2. A monotonic exponential $e^{\sigma t}$ $(\omega = 0, s = \sigma)$ …(fig. (a))
3. A sinusoid $\cos \omega t$ $(\sigma = 0, s = \pm j\omega)$ …(fig. (b))
4. An exponentially varying sinusoid $e^{\sigma t} \cos \omega t$ $(s = \sigma \pm j\omega)$ …(fig. (c and d))

There functions are illustrated in following figures.



Sinusoids of complex frequency $\sigma + j\omega$

The complex frequency s can be conveniently represented on a complex frequency plane (s plane) as depicted in following figure. The absolute value of the imaginary part of s is $|\omega|$ (the radian frequency), which indicates the frequency of oscillation of e^{st}; the real part σ (the neper frequency) gives information about the rate of increase or decrease of the amplitude of e^{st}.



Complex frequency plane(s-plane)

Note: Thus the s-plane can be differentiated into two parts: *the left half-plane (LHP)* corresponding to exponentially decaying signals and the *right half-plane (RHP)* corresponding to exponentially growing signals. The *imaginary axis* separates the two regions and corresponds to signals of constant amplitude.

Even and Odd Signals

A real valued signal f(t) is called symmetric (even) if

$$x(-t) = f(t)$$

and is called anti–symmetric (odd) if

$$x(-t) = -f(t)$$

For an odd function x(0) = 0

Even and Odd Components of signals

Every signal f(t) can be expressed as a sum of even and odd components

i.e. $f(t) = f_e(t) + f_o(t)$

where $f_e(t)$ is the even part of f(t)

$f_o(t)$ is the odd part of f(t)

The even part $x_e(t) = ½ [f(t) + f(-t)]$

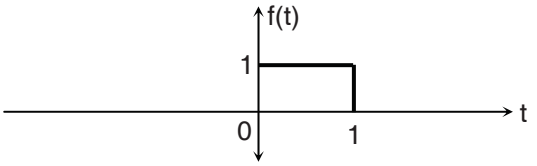
The odd part $x_o(t) = ½ [f(t) - f(-t)]$

For odd function, f(0) = 0 always

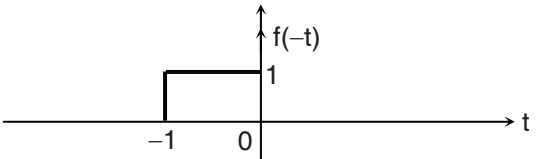
Example:

1. Find the even/odd part of the signal.

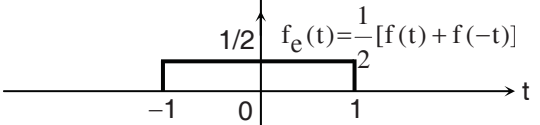
Solution :



Step (i)

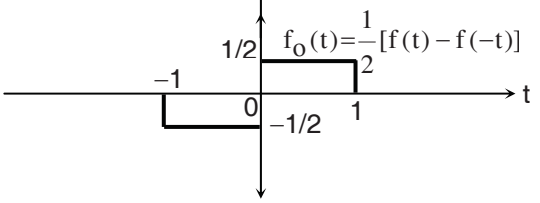


Step (ii)



Even component

Step (iii)



Odd component

Some useful properties of even and odd function

(a) *$f_e(t)$ and $f_o(t)$ are orthogonal functions.*

i.e. $$\int_{-\infty}^{\infty} f_e(t) f_o(t)\,dt = 0 .$$

(b) *Energy of signal f(t) is equal to*

$$E = \int_{-\infty}^{\infty} |f(t)|^2\,dt = \int_{-\infty}^{\infty} f_e^2(t)\,dt + \int_{-\infty}^{\infty} f_o^2(t)\,dt$$

Proof :

$$E = \int_{-\infty}^{\infty} f^2(t).dt = \int_{-\infty}^{\infty} [f_e(t)+f_o(t)]^2 .dt$$

$$= \int_{-\infty}^{\infty} [f_e^2(t) + 2f_e(t).f_o(t) + f_o^2(t)].dt$$

$$= \int_{-\infty}^{\infty} f_e^2(t)\,dt + 2\int_{-\infty}^{\infty} f_e(t).f_o(t).dt + \int_{-\infty}^{\infty} f_o^2(t).dt = \int_{-\infty}^{\infty} f_e^2(t) + \int_{-\infty}^{\infty} f_o^2(t).dt$$

$\because \int_{-\infty}^{\infty} f_e(t).f_o(t).dt = 0$ as $f_e(t)$ and $f_o(t)$ are orthogonal Thus

$$E = \int_{-\infty}^{\infty} f_e^2(t)\,dt + \int_{-\infty}^{\infty} f_o^2(t)\,dt$$

(c) *Area*

Because $f_e(t)$ is symmetrical about the vertical axis, it follows that

$$\int_{-a}^{a} f_e(t)dt = 2\int_{0}^{a} f_e(t)dt$$

It is also clear that $\int_{-a}^{a} f_0(t)dt = 0$

(d) even function × odd function = odd function
odd function × odd function = even function
even function × even function = even function

CLASSIFICATION OF SYSTEMS

Systems may be classified broadly in the following categories :

1. Linear and Non-linear Systems

A system whose output is proportional to its input is an example of a linear system. But linearity implies more than this; it also implies *additivity property*. i.e. if a cause c_1 acting alone has an effect e_1, and if another cause c_2, also acting alone, has an effect e_2, then, with both causes acting on the system, the total effect will be $e_1 + e_2$. Thus, if

$$c_1 \rightarrow e_1 \quad \text{and} \quad c_2 \rightarrow e_2$$

then for all c_1 and c_2

$$c_1 + c_2 \rightarrow e_1 + e_2$$

In addition, a linear system must satisfy the *homogeneity* or *scaling* property, i.e. if a cause is increased k-fold, the effect also increases k-fold. Thus, if

$$c \rightarrow e$$

then for all real or imaginary k

$$kc \rightarrow ke$$

Both these properties can be combined into one property (*superposition*), which is expressed as follows : if

$$c_1 \rightarrow e_1 \quad \text{and} \quad c_2 \rightarrow e_2$$

then for all values of constants k_1 and k_2,

$$k_1c_1 + k_2c_2 \rightarrow k_1e_1 + k_2e_2$$

This is true for all c_1 and c_2.

Note : Almost all systems observed in practice becomes non-linear when large enough signals are applied to them. However, many systems show linear behavior for small signals or can be linearzed for small signal condition using Taylor series expansion about the operating point.

2. Time-Invariant and Time-Varying Parameter Systems :

Systems whose parameters do not change with time are time-invariant (also constant-parameter) systems. For such a system, if the input is delayed by T seconds, the output is the same as before but delayed by T.
A system with an input-output relationship described by a linear differential equation with constant coefficients is a linear time-invariant (LTI) system. If these coefficients are functions of time, then the system is a linear time-varying system. *eg. Rocket*.

3. Instantaneous and Dynamic Systems:

If output of the network at some instant t depends only on the input at that instant t, then such systems are said to be instantaneous or memory less systems. i.e. A system is said to be instantaneous (or memory less) if its output at any instant t depends, at most, on the strength of its inputs at the same instant but not on any past or future values of the inputs. Otherwise, the system is said to be dynamic (or a system with memory). Networks containing inductive and capacitive elements generally have infinite memory.

4. Causal and Noncausal Systems :

A causal (also known as a physical or non-anticipative) system is one for which the output at any instant t_0 depends only on the value of the inputs f(t) for $t \leq t_0$. In order words, the value of the output at the present instant depends only on the past and present values of the inputs f(t), not on its future values. If the response starts before the input, it means that the system knows the input in the future and acts on this knowledge before the input is applied. Such systems are called *non-causal (or anticipative)* system. Any practical system that operates in real time must necessarily be causal.

Note : It is impossible to implement non-causal system in real time.

5. Lumped-parameter and Distributed-Parameter Systems :

If the physical dimensions of a component are small compared to the wavelength of the signal propagated, we may assume that the current is constant throughout the component. This is the assumption made in *lumped-parameter systems*, where each component is regarded as being lumped at one point in space. Such an assumption is justified at lower frequencies (higher wavelength). For such systems, the system equations require only one independent variable (time) and therefore are described by ordinary differential equations.

In contrast, for *distributed-parameter systems* such as transmission lines, waveguides, antennas, and microwave tubes, the system dimensions cannot be assumed to be small compared to the wavelengths of the signals; thus the lumped-parameter assumption breaks down. The signals here are functions of space as well as of time, leading to mathematical models consisting of partial differential equations.

6. Continuous-Time and Discrete-Time Systems :

Systems whose inputs and outputs are continuous-time signals are continuous-time systems. On the other hand, systems whose inputs and outputs are discrete-time signals are discrete-time systems. If a continuous-time signal is sampled, the resulting signal is a discrete-time signal. We can process a continuous-time signal by processing its samples with a discrete-time system.

7. Analog and Digital Systems :

A system whose input and output signals are analog is an analog system; a system whose input and output signals are digital is a digital system. A digital computer is an example of a digital (binary) system. Observe that a digital computer is an example of a system that is digital as well as discrete-time.

Examples:

1. Determine whether following systems are linear or non-linear

(i) $3y(t) + 2 = f(t)$ (ii) $\frac{dy}{dt} + 3ty(t) = t^2 f(t)$

Solution :

(i) $3y(t) + 2 = f(t)$

if inputs $f_1(t)$ and $f_2(t)$ are applied separately , we have

$3y_1(t) + 2 = f_1(t)$ …(5)

$3y_2(t) + 2 = f_2(t)$ …(6)

multiplying (5) by a and (6) by b and adding we get,

$3[ay_1(t) + by_2(t)] + 2(a+b) = [af_1(t) + bf_2(t)]$ …(7)

if both input $af_1(t)$ and $bf_2(t)$ are applied simultaneously, we must get,

$3[ay_1(t) + by_2(t)] + 2 = [af_1(t) + bf_2(t)]$ …(8)

we see that equation (7) $\neq$ equation (8)
$\therefore$ given system is non-linear,

Note: The above system can be linearized if constant '2' is abstent.

(ii) $\frac{dy}{dt} + 3ty(t) = t^2 f(t)$

if inputs $f_1(t)$ and $f_2(t)$ are applied separately , we have

$$\frac{dy_1}{dt} + 3ty_1(t) = t^2 f_1(t) \quad \text{…(9)}$$

$$\frac{dy_2}{dt} + 3ty_2(t) = t^2 f_2(t) \quad \text{…(10)}$$

multiplying (9) by a and (10) by b and adding we get,

$$\frac{d}{dt}[ay_1(t) + by_2(t)] + 3t[ay_1(t) + by_2(t)] = t^2[af_1(t) + bf_2(t)] \quad \text{…(11)}$$

if both input $af_1(t)$ and $bf_2(t)$ are applied simultaneously, we must get,

$$\frac{d}{dt}[ay_1(t) + by_2(t)] + 3t[ay_1(t) + by_2(t)] = t^2[af_1(t) + bf_2(t)] \quad \text{…(12)}$$

we see that equation (11) = equation (12) $\therefore$ given system is linear.

2. Determine whether following systems are time invariant or time varying

(i) $y(t) = f(t-2)$ (ii) $y(t) = tf(t-2)$

Solution:

(i) $y(t) = f(t-2)$

let the input be delayed by T second, then

$y_1(t) = f(t-T-2)$ …(13)

now let the output y(t), be delayed by T seconds, then

$y_2(t) = y(t-T) = f(t-T-2)$ …(14)

$\therefore$ (13) = (14)

i.e. $y_1(t) = y_2(t)$

system is time invariance

(ii) $y(t) = tf(t-2)$

let the input be delayed by T seconds, then

$y_1(t) = t\,f(t-T-2)$ …(15)

now, let y(t) be delayed by T seconds, then

$y_2(t) = y(t-T)$

$= (t-T)\,f(t-T-2)$ …(16)

here $y_1(t) \neq y_2(t)$

$\therefore$ system is time varying

TIME–DOMAIN ANALYSIS OF LTIC SYSTEMS

Introduction

Consider linear differential systems with constant co-efficient for which the input f(t) and the output y(t) are related by linear differential equations of the form

$$\frac{d^n y}{dt^n} + a_{n-1}\frac{d^{n-1}y}{dt^{n-1}} + \ldots + a_1\frac{dy}{dt} + a_0 y(t) = b_m\frac{d^m f}{dt^m} + b_{m-1}\frac{d^{m-1}f}{dt^{m-1}} + \ldots + b_1\frac{df}{dt} + b_0 f(t) \quad \ldots(17)$$

Using operational notation D to represent d/dt, we can express this equation as

$$(D^n + a_{n-1}D^{n-1} + \ldots + a_1 D + a_0)y(t) = (b_m D^m + b_{m-1}D^{m-1} + \ldots + b_1 D + b_0)f(t)$$

or $Q(D)y(t) = P(D)f(t)$

where the polynomials Q(D) and P(D) are

$$Q(D) = D^n + a_{n-1}D^{n-1} + \ldots + a_1 D + a_0$$

$$P(D) = b_m D^m + b_{m-1}D^{m-1} + \ldots + b_1 D + b_0$$

Note : Practical noise considerations, requires $m \le n$.

A system described by Equation (17) is linear and its response can be expressed as the sum of two components: the zero-input component and the zero-state component.

Therefore,

Total response = zero-input response + zero-state response

The zero-input component is the system response when the input f(t) = 0 so that it is the result of internal system conditions (such as energy storages, initial conditions) alone. It is independent of the external input f(t). In contrast, the zero-state component is the system response to the external input f(t) when the system is in zero state, that is, all initial conditions are zero.

System Response to Internal Conditions: Zero-Input Response

The zero-input response $y_0(t)$ is the solution of equation (17) when the input f(t) = 0 so that

$$Q(D)y_0(t) = 0$$

or $$(D^n + a_{n-1} D^{n-1} + \ldots + a_1 D + a_0)\, y_0(t) = 0 \qquad \ldots(18)$$

Equation (18) shows that a linear combination of $y_0(t)$ and its n successive derivatives is zero, not at some values of t, but for all t. Such a result is possible if and only if $y_0(t)$ and all its n successive derivatives are of the same form. Only an exponential function $e^{\lambda t}$ has this property. So let us assume that

$$y_0(t) = ce^{\lambda t}$$

Substituting these results in equation (18), we obtain

$$c(\lambda^n + a_{n-1}\lambda^{n-1} + \ldots + a_1\lambda + a_0)e^{\lambda t} = 0$$

For a nontrivial solution of above equation,

$$\lambda^n + a_{n-1}\lambda^{n-1} + \ldots + a_1\lambda + a_0 = 0 \qquad \ldots(19)$$

This result means that $ce^{\lambda t}$ is indeed a solution of equation (18), provided that λ satisfies equation (19). Note that the polynomial in equation (19) is identical to the polynomial Q(D) in equation (18), with λ replacing D. Therefore, equation (19) can be expressed as

$$Q(\lambda) = 0 \qquad \ldots(20)$$

When $Q(\lambda)$ is expressed in factorized form, equation (20) can be represented as

$$Q(\lambda) = (\lambda - \lambda_1)(\lambda - \lambda_2)\cdots(\lambda - \lambda_n) = 0$$

We can readily show that a general solution is given by the sum of these n solutions, so that

$$y_0(t) = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t} + \cdots + c_n e^{\lambda_n t}$$

where c_1, c_2, … , c_n are arbitrary constants determined by n constraints (the initial conditions) on the solution.

The equation

$$Q(\lambda) = 0$$

is called the *characteristic* equation of the system. λ_1, λ_2, …, λ_n are the roots of the characteristic equation; consequently, they are called the *characteristic roots, characteristic values, eigenvalues*, and *natural frequencies*. The exponentials $e^{\lambda_i t}(i = 1,2,....,n)$ in the zero-input response are the *characteristic modes* (also known as *modes* or *natural modes*) of the system. The single most important attribute of an LTIC system is its characteristic modes. Characteristic modes not only determine the zero-input response but also play an important role in determining the zero-state response.

The zero input response, $y_0(t)$ depends upon the nature of the characteristic roots of equation $Q(\lambda) = 0$.

Nature of roots of $Q(\lambda) = 0$	Zero-input response, $y_0(t)$
1. All distinction roots	$y_0(t) = c_1e^{\lambda_1 t} + c_2e^{\lambda_2 t} + \cdots c_ne^{\lambda_n t}$
2. Some roots repeated (say r-times)	$y_0(t) = (c_1 + c_2t + c_3t^2 + \cdots + c_rt^{r-1})e^{\lambda t}$
3. Complex conjugate roots of the form $\alpha \pm j\beta$	$y_0(t) = ce^{\alpha t}\cos(\beta t + \theta)$

Example

1. Find $y_0(t)$, the zero-input component of the response for an LTI system described by the following differential equation :

$$(D^2 + 3D + 2)y(t) = Df(t)$$

when the initial condition are $y_0(0) = 0, \dot{y}_0(0) = -5$.

Solution:

Note that $y_0(t)$, being the zero-input component (f(t) = 0), is the solution of $(D^2 + 3D + 2)$ $y_0(t) = 0$. The characteristic polynomial of the system is $\lambda^2 + 3\lambda + 2 = 0$. The characteristic roots of the system are $\lambda_1 = -1$ and $\lambda_2 = -2$. Consequently, the zero-input response is

$$y_0(t) = c_1e^{-t} + c_2e^{-2t} \qquad \ldots(21)$$

To determine the arbitrary constants c_1 and c_2, we differentiate equation (21) to obtain

$$\dot{y}_0(t) = c_1e^{-t} - 2c_2e^{-2t} \qquad \ldots(22)$$

Setting t = 0 in equations (21) and (22), and substituting the initial conditions $y_0(0) = 0$ and $\dot{y}_0(0) = -5$ we obtain

$$0 = c_1 + c_2$$
$$-5 = -c_1 - 2c_2$$

Solving these two simultaneous equations in two unknowns for c_1 and c_2 yields

$$c_1 = -5, \qquad c_2 = 5$$

Therefore $y_0(t) = -5e^{-t} + 5e^{-2t}, \qquad \text{for } t \geq 0.$

2. Similar procedure may be followed for repeated roots. For instance, for a system specified by

$$(D^2 + 6D + 9)y(t) = (3D + 5)f(t)$$

let us determine $y_0(t)$, the zero-input component of the response if the initial onditions are

$$y_0(0) = 3 \text{ and } \dot{y}_0(0) = -7$$

Solution:

The characteristic polynomial is $\lambda^2 + 6\lambda + 9 = (\lambda + 3)^2 = 0$, and its characteristic roots are $\lambda_1 = -3$, $\lambda_2 = -3$ (repeated roots). The zero-input response is given by

$$y_0(t) = (c_1 + c_2 t)e^{-3t}$$

We can find the arbitrary constants c_1 and c_2 from the initial conditions $y_0(0) = 3$ and $\dot{y}_0(0) = -7$ following the same procedure in example (1). Which gives $c_1 = 3$ and $c_2 = 2$. Hence,

$$y_0(t) = (3 + 2t)e^{-3t}, \qquad t \geq 0$$

3. For the case of complex roots, let us find the zero-input response of an LTI system described by the equation:

$$(D^2 + 4D + 40)\, y(t) = (D + 2)f(t)$$

with initial conditions $y_0(0) = 2$ and $\dot{y}_0(0) = 16.78$

Solution:

The characteristic polynomial is $\lambda^2 + 4\lambda + 40 = (\lambda + 2 - j6)\ (\lambda + 2 + j6) = 0$. The characteristic roots are $-2 \pm j6$. Since $\alpha = -2$ and $\beta = 6$, the real form solution is [see point 3 in the above table]

$$y_0(t) = ce^{-2t}\cos(6t + \theta) \qquad \ldots(23)$$

where c and θ are arbitrary constants to be determined from the initial conditions $y_0(0) = 2$ and $\dot{y}_0(0) = 16.78$. Differentiation of equation (23) yields

$$\dot{y}_0(t) = -2ce^{-2t}\cos(6t + \theta) - 6ce^{-2t}\sin(6t + \theta) \qquad \ldots(24)$$

Setting t = 0 in equations (23) and (24), and then substituting initial conditions, we obtain

$$2 = c\cos\theta$$
$$16.78 = -2c\cos\theta - 6c\sin\theta$$

Solution of these two simultaneous equation, we get

$$c \cos \theta = 2 \qquad \ldots(25)$$

$$c \sin \theta = -3.463 \qquad \ldots(26)$$

Squaring and then adding the two sides of the above equations yields

$$c^2 = (2)^2 + (-3.464)^2 = 16 \Rightarrow c = 4$$

Next, dividing equation (26) by (25), we get

and $$\tan\theta = \frac{-3.463}{2}$$

$$\theta = \tan^{-1}\left(\frac{-3.463}{2}\right) = -\frac{\pi}{3}$$

Therefore $$y_0(t) = 4e^{-2t}\cos\left(6t - \frac{\pi}{3}\right)$$

Practical Initial Conditions and the meaning of 0^- and 0^+

The input is assumed to start at t = 0, unless otherwise mentioned. Hence, t = 0 is the reference point. The conditions immediately before t = 0 (just before the input is applied) are the conditions at $t = 0^-$, and those immediately after t = 0 (just after the input is applied) are the conditions at $t = 0^+$ (compare this with the historical time frame B.C. and A.D.). In practice, we are likely to know the initial conditions at $t = 0^-$ rather than at $t = 0^+$. The two sets of conditions are generally different, although in some cases they may be identical.

We are dealing with the total response y(t), which consists of two components; the zero-input component $y_0(t)$ (response due to the initial conditions alone with f(t) = 0) and the zero-state component resulting from the input alone with all initial conditions zero. At $t = 0^-$, the response y(t) consists solely of the zero-input component $y_0(t)$ because the input has not started yet. Hence the initial conditions on y(t) are identical to those of $y_0(t)$. Thus, $y(0^-) = y_0(0^-)$, $\dot{y}(0^-) = \dot{y}_0(0^-)$, and so on. Moreover, $y_0(t)$ is the response due to initial conditions alone and does not depend on the input f(t). Hence, application of the input at t = 0 does not affect $y_0(t)$. This means the initial conditions on $y_0(t)$ at t = 0 and 0^+ are identical; that is $y_0(0^-)$, $\dot{y}_0(0^-), \ldots$ are identical to $y_0(0^+), \dot{y}_0(0^+), \ldots$, respectively. It is clear that for $y_0(t)$, there is no distinction between the initial conditions at $t = 0^-$, 0 and 0^+. They are all the same. But this is not the case with the total response y(t), which consists of both, the zero-input and the zero-state components. Thus, in general, $y(0^-) \neq y(0^+)$, $\dot{y}(0^-) \neq \dot{y}(0^+)$, and so on.

The Unit Impulse Response h(t)

h(t) is the system response to an impulse input $\delta(t)$ applied at t = 0 with all the initial conditions zero at $t = 0^-$. Impulse input $\delta(t)$ appears momentarily at t = 0, and then it is gone forever. But in that moment it generates energy storages; that is, it creates nonzero initial conditions instantaneously within the system at $t = 0^+$. Although the impulse input $\delta(t)$ vanishes for t > 0 so that the system has no input after the impulse has been applied, the system will still have a response generated by these newly created initial conditions.

Thus impulse response h(t), therefore, must consist of the system's characteristic modes for $t \geq 0^+$.

$$h(t) = \text{characteristic mode terms} \qquad t \geq 0^+$$

This response is valid for t > 0. But what happens at t = 0? At a single moment t = 0, there can at most be an impulse, so the form of the complete response h(t) is given by

$$h(t) = A_0\delta(t) + \text{characteristic mode terms} \qquad t \geq 0$$

which takes the form

$$h(t) = b_n\delta(t) + [P(D)y_n(t)u(t)]$$

where b_n is the coefficient of the nth-order term in P(D), and $y_n(t)$ is a linear combination of the characteristic modes of the system subject to the following initial conditions:

$$y_n^{(n-1)}(0) = 1, \text{ and } \quad y_n(0) = \dot{y}_n(0) = \ddot{y}_n(0) = \cdots = y_n^{(n-2)}(0) = 0$$

Note : Once we know the system response to an impulse input, we can determine the system response to an arbitrary input f(t).

Example:

1) Find unit impulse response of LTIC system described by equation
(D + 2) y(t) = (3D + 5) f(t)

Solution:

Here Q(D) = D + 2 and P(D) = 3D + 5 and also $b_n = b_1 = 3$ the characteristic equation is

$$Q(\lambda) = 0 \quad \Rightarrow \quad \lambda + 2 = 0$$
$$\Rightarrow \quad \lambda = -2$$

$\therefore \quad y_n(t) = Ce^{-2t}$ subject to $y_n(0) = 1$

$\Rightarrow \quad C = 1$

$\therefore \quad y_n(t) = e^{-2t}$

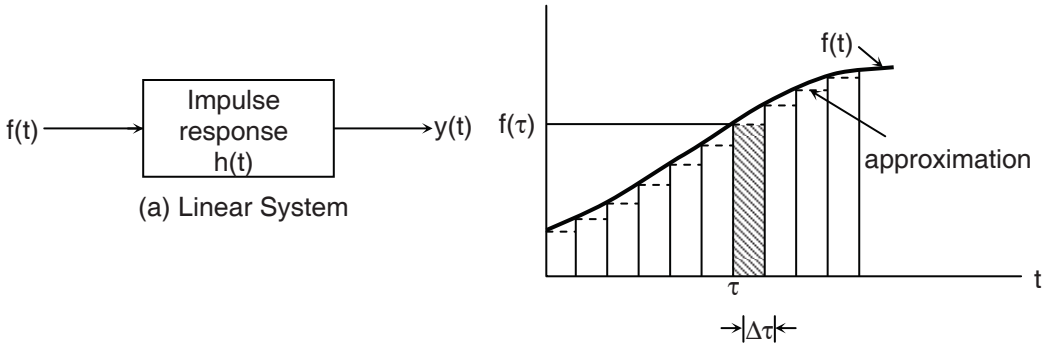
Impulse response

$$\begin{aligned}
\therefore \quad h(t) &= b_n\,\delta(t) + P(D)\,[y_n(t)]\,u(t) \\
&= 3\,\delta(t) + [(3D + 5)\,e^{-2t}]\,u(t) \\
&= 3\,\delta(t) + (5e^{-2t} - 6e^{-2t})\,u(t) \\
&= 3\delta(t) - e^{-2t}\,u(t)
\end{aligned}$$

thus, $\quad h(t) = 3\delta(t) - e^{-2t}\,u(t)$

Signal Transmission through LTI Systems

In the time domain, a linear system is described in terms of its impulse response, which is defined as the response of the system (with zero initial conditions) to a unit impulse or delta function $\delta(t)$ applied to the input of the system. If the system is time-invariant, then the shape of the impulse response is the same no matter when the unit impulse is applied to the system. Let us denote the impulse response of a linear time-invariant system by h(t). Let this system be subjected to an arbitrary excitation f(t), as in following figure (a).



(a) Linear System

(b) Approximation of input f(t)

To determine the response y(t) of the system, we begin by first approximating f(t) by a staircase function composed of narrow rectangular pulses, each of duration $\Delta\tau$, as shown in above figure (b).

Clearly in the limit this approachest, a delta function weighted by a factor equal to the height of the pulse times $\Delta\tau$. Consider a typical pulse, shown shaded in above figure (b), which occurs at $t = \tau$. This pulse has an area equal to $f(\tau)\Delta\tau$.

By definition, the response of the system to a unit impulse or delta function $\delta(t)$, occurring at t = 0, is h(t). It follows, therefore, that the response of the system to a delta function, weighted by the factor $f(\tau)\Delta\tau$ and occurring at $t = \tau$, must be $f(\tau)h(t-\tau)\Delta\tau$. To find the total response y(t) at some time t, we apply the principle of superposition. Thus, summing the various infinitesimal responses due to the various input pulses, we obtain in the limit, as $\Delta\tau$ approaches zero.

$$y(t) = \int_{-\infty}^{\infty} f(\tau)h(t-\tau)\,d\tau \qquad \ldots(27)$$

This relation is called the *convolution integral*.

In equation (27), three different time scales are involved: excitation time τ, response time t, and system-memory time $t - \tau$. This relation is the basis of time-domain analysis of linear time-invariant systems. It states that the present value of the response of a linear time-invariant system is a weighted integral over the past history of the input signal, weighted according to the impulse response of the system. Thus the impulse response acts as a memory function for the system.

System Response to External Input : Zero State Response

This is the system response y(t) to an input f(t) when the system is in zero state; that is, when all initial conditions are zero. Under these conditions, the zero-state response will be the total response of the system.

The (zero-state) response y(t) to the input f(t) is given by

$$y(t) = \int_{-\infty}^{\infty} f(\tau)h(t-\tau)d\tau \qquad \ldots(28)$$

Above equation shows that knowing h(t), we can determine the response y(t) to any input. Observe the all-pervasive nature of the system's characteristic modes. The system response to any input is determined by the impulse response, which in turn is made up of characteristic modes of the system. The above equation (28) is only valid for LTIC systems.

Convolution and Its Graphic Interpretation

(i) *Convolution Integral*

The zero-state response y(t) is given as

$$y(t) = \int_{-\infty}^{\infty} f(\tau)h(t-\tau)d\tau \qquad \ldots(29)$$

Equation (29) occurs frequency in the physical sciences, engineering, and mathematics. For this reason this integral is given a special name: the *convolution integral*. The convolution integral of two function $f_1(t)$ and $f_2(t)$ is denoted symbolically by $f_1(t) * f_2(t)$ and is defined as

$$f_1(t) * f_2(t) \equiv \int_{-\infty}^{\infty} f_1(\tau)f_2(t-\tau)\,d\tau$$

Properties :

1) *Commutative* : $x_1(t) * x_2(t) = x_2(t) * x_1(t)$

2) *Associative* : $[x_1(t) * x_2(t)] * x_3(t) = x_1(t) * [x_2(t) * x_3(t)]$

3) *Distributive* : $x_1(t) * [x_2(t) + x_3(t)] = [x_1(t) * x_2(t) + x_1(t) * x_3(t)]$

4) *The Shift Property* :

If $f_1(t) * f_2(t) = c(t)$

then $f_1(t) * f_2(t-T) = c(t-T)$

$f_1(t-T) * f_2(t) = c(t-T)$

and

$f_1(t-T_1) * f_2(t-T_2) = c(t-T_1-T_2)$

5) *Convolution with an Impulse*:

Convolution of a function f(t) with a unit impulse results in the function f(t) itself. Therefore

$f(t) * \delta(t) = f(t)$

6) *The Width Property* :

If the durations (widths) of $f_1(t)$ and $f_2(t)$ are T_1 and T_2 respectively, then the duration (width) of $f_1(t) * f_2(t)$ is $(T_1 + T_2)$

Note: In deriving equation (29), we assumed the system to be LTI. There were no other restrictions either on the system or on the input signal f(t). In practice, most systems are causal, so that their response cannot begin before the input starts. Furthermore, most inputs are also causal, which means they start at t = 0.

Causality restrictions on both signals and systems further simplify the limits of integration in equation (29). It is important to remember that the integration in equation (29) is performed with respect to τ (not t). If the input f(t) is causal, $f(\tau) = 0$ for $\tau < 0$. Therefore, $f(\tau) = 0$ for $\tau < 0$. Similarly, if h(t) is causal, $h(t - \tau) = 0$ for $t - \tau < 0$; that is, for $\tau > t$. Therefore, the product $f(\tau)h(t-\tau) = 0$ everywhere except over the interval $0 \le \tau \le t$. Therefore equation (29) reduces to

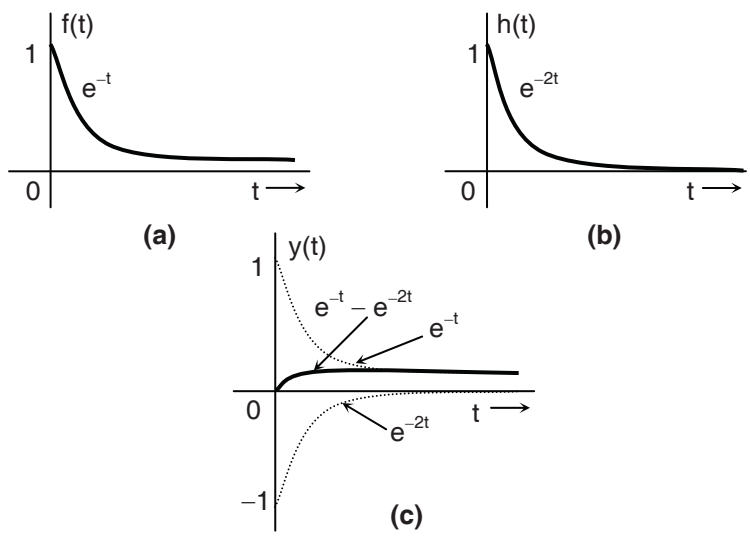
$$y(t) = f(t) * h(t) = \int_{0-}^{t} f(\tau)h(t-\tau)d\tau \qquad t \ge 0 \qquad \ldots(30)$$
$$= 0 \qquad t < 0$$

The lower limit of integration in equation (30) is taken as 0^- to avoid the difficulty in integration that can arise if f(t) contains an impulse at the origin.

Example

For an LTIC system with the unit impulse response $h(t) = e^{-2t}\,u(t)$, determine the response y(t) for the input

$f(t) = e^{-t}u(t)$



Convolution of f(t) and h(t) in above example

Here both f(t) and h(t) are causal (see above figure). Hence, we need only to perform the convolution's integration over the range (0, t) [see equation (30)]. The system response is therefore given by

$$y(t) = \int_0^t f(\tau)h(t-\tau)\,d\tau \qquad t \ge 0$$

$$y(t) = \int_0^t e^{-\tau}e^{-2(t-\tau)}d\tau \qquad t \ge 0$$

$$y(t) = e^{-2t}\int_0^t e^{\tau}d\tau = e^{-2t}(e^t - 1) = e^{-t} - e^{-2t} \qquad t \ge 0 \qquad \ldots(31)$$

Also, y(t) = 0 when t < 0 [see equation (30)]. This result, along with equation (31), yields

$$y(t) = (e^{-t} - e^{-2t})u(t)$$

The response is depicted in above figure (c).

(ii) *Graphical Interpretation of convolution*

To have a proper grasp of convolution operation, we should understand the graphical interpretation of convolution. In addition, graphical convolution allows us to grasp visually or mentally the convolution integral's result, which can be of great help in sampling, filtering, and many other problems.

The procedure for graphical convolution can be summarized as follows:

1. Keep the function $f(\tau)$ fixed.
2. Visualize the function $g(\tau)$ as a rigid wire frame, and rotate (or invert) this frame about the vertical axis ($\tau = 0$) to obtain $g(-\tau)$.
3. Shift the inverted frame along the τ axis by t_0 seconds. The shifted frame now represents $g(t_0 - \tau)$.
4. The area under the product of $f(\tau)$ and $g(t_0 - \tau)$ (the shifted frame) is $c(t_0)$, the value of the convolution at $t = t_0$.
5. Repeat this procedure, shifting the frame by different values (positive and negative) to obtain c(t) for all values of t.

Example

Find c(t) = f(t) * g(t) for the signals depicted in following figures (a) and (b). Since f(t) is simpler than g(t), it is easier to evaluate g(t) * f(t) than f(t) * g(t). However, we have intentionally take the more difficult route and evaluate f(t) * g(t) to clarity some of the finer points of convolution.

Following figure (a) and (b) shows f(t) and g(t) respectively. Observe that g(t) is composed of two segments. As a result, it can be described as

$$g(t) = \begin{cases} 2e^{-t} & \text{segment A, } t > 0 \\ -2e^{2t} & \text{segment B, } t < 0 \end{cases}$$

$$g(t-\tau) = \begin{cases} 2e^{-(t-\tau)} & \text{segment A, } \tau < \tau \\ -2e^{2(t-\tau)} & \text{segment B, } \tau > t \end{cases}$$

The segment of f(t) that is used in convolution is f(t) = 1, so that $f(\tau) = 1$. Following figure (c) shows $f(\tau)$ and $g(-\tau)$.

To compute c(t) for $t \geq 0$, we right-shift $g(-\tau)$ to obtain $g(t - \tau)$, as illustrated in following figure (d). Clearly, $g(t - \tau)$ overlaps with $f(\tau)$ over the shaded interval; that is, over the range $\tau \geq 0$; Segment A overlaps with $f(\tau)$ over the interval (0, t), while Segment B overlaps with $f(\tau)$ over (t, ∞). Remembering that $f(\tau) = 1$, we have

$$\begin{aligned} c(t) &= \int_0^{\infty} f(\tau)g(t-\tau)d\tau \\ &= \int_0^{t} 2e^{-(t-\tau)}d\tau + \int_t^{\infty} -2e^{2(t-\tau)}d\tau = 2(1-e^{-t}) - 1 \\ &= 1 - 2e^{-t} \qquad t \geq 0 \end{aligned}$$

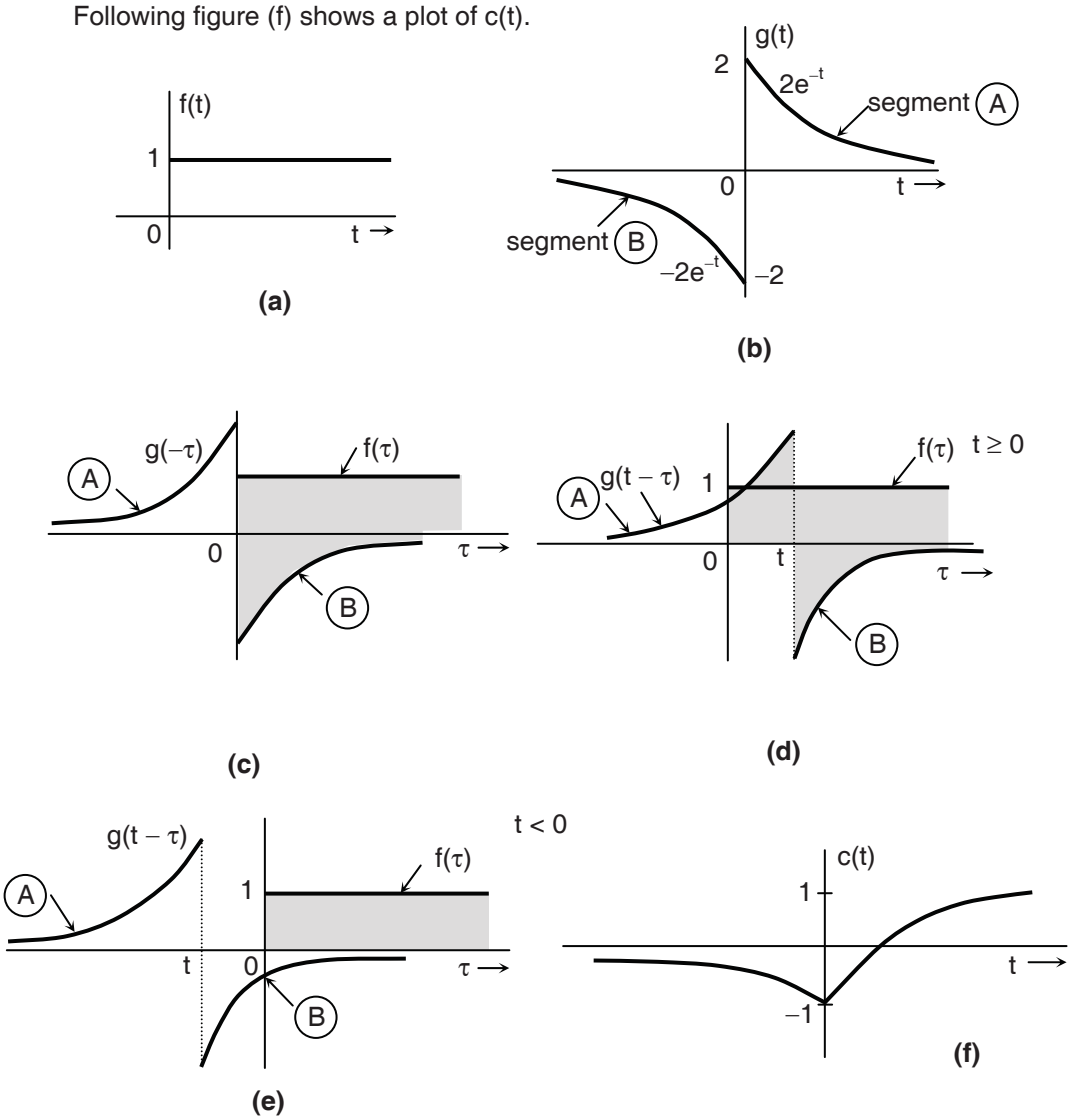
Following figure (e) shows the situation for t < 0. Here the overlap is over the shaded interval; that is, over the range $\tau \geq 0$, where only the segment B of g(t) is involved. Therefore

$$c(t) = \int_0^{\infty} f(\tau)g(t-\tau)d\tau = \int_0^{\infty} g(t-\tau)d\tau = \int_0^{\infty} -2e^{2(t-\tau)}\,d\tau$$

$$= -e^{2t} \qquad t < 0$$

$$c(t) = \begin{cases} 1-2e^{-2t} & t \geq 0 \\ -e^{2t} & t < 0 \end{cases}$$

Following figure (f) shows a plot of c(t).



Convolution of f(t) and g(t)

Total Response

The total response of a linear system can be expressed as the sum of its zero-input and zero-state components:

$$\text{Total Response} = \underbrace{\sum_{j=1}^{n} c_j e^{\lambda_j t}}_{\text{zero-input component}} + \underbrace{f(t) * h(t)}_{\text{zero-state component}}$$

Example:

1. Find the total response y(t) for the equation

$$\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y(t) = \frac{df}{dt}$$

if the input $f(t) = 10\, e^{-3t}\, u(t)$ and $y(0^-) = 0$ and $\dot{y}(0^-) = -5$

Solution:

[$\because$ initial condution are given at $t = 0^-$, if applies to zero-input response i.e. $y_0(0) = y(0^-)$ and $\dot{y}(0^-)$]

i) zero-input response, $y_0(t)$

The given differential equation is

$$\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y(t) = \frac{df}{dt}$$

using 'D' operator, we get

$$\left(D^2 + 3D + 2\right) y(t) = Df(t)$$

to find zero-input response we will consider f(t) = 0

$\therefore \quad (D^2 + 3D + 2)\, y_0(t) = 0$

$\therefore$ characteristic equation is

$$\lambda^2 + 3\lambda + 2 = 0$$

$\Rightarrow \quad (\lambda + 2)(\lambda + 1) = 0$

$\Rightarrow \quad \lambda_1 = -1$ and $\lambda_2 = -2$

$\therefore \quad y_0(t) = C_1 e^{-t} + C_2 e^{-2t}$

$\dot{y}_0(t) = -C_1 e^{-t} - 2C_2 e^{-2t}$

Putting the values $y_0(0) = 0$ and $\dot{y}_0(0) = -5$, we get

$C_1 + C_2 = 0$

$C_1 + 2C_2 = 5$ …solving we get $C_1 = -5$ and $C_2 = 5$

$\therefore \quad y_0(t) = -5e^{-t} + 5e^{-2t} \quad \rightarrow$ required zero-input response

ii) To find zero-state response, we first have to find impilse response h(t), will be of the form shown below

$\therefore \quad h(t) = b_n\, \delta(t) + P(D)\ [y_n(t)]\, u(t)$

where $b_n = b_2 = 0$ and $y_n(t) = k_1e^{-t} + k_2e^{-2t}$ subject to $\dot{y}_n(0) = 1, y_n(0) = 0$

$\Rightarrow \quad k_1 + k_2 = 0$

$\quad - \quad k_1 - 2k_2 = 1$ … solving we get $k_1 = 1$, $k_2 = -1$

$\therefore \quad y_n(t) = e^{-t} - e^{-2t}$

$\therefore \quad h(t) = P(D)\,[y_n(t)]\,u(t)$

$\quad = D[e^{-t} - e^{-2t}]u(t)$

$h(t) = (-e^{-t} + 2e^{-2t})\,u(t)$ impulse response

iii) zero-state response, $y_{zs}(t) = h(t) * f(t) = f(t) * h(t)$

$$= f(t) * [-e^{-t}\,u(t) + 2e^{-2t}\,u(t)]$$

$$= -f(t) * e^{-t}\,u(t) + 2\,f(t) * e^{-2t}\,u(t)$$

$$= -\int_0^t 10e^{-3t}.e^{-(t-\tau)}d\tau + 2\int_0^t 10e^{-3\tau}.e^{-2(t-\tau)}d\tau$$

…[$\because$ both f(t) and h(t) are causal, limits are form $0 \to t$]

$$= -10e^{-t}\int_0^t e^{-2\tau}d\tau + 20e^{-2\tau}\int_0^t e^{-\tau}\,d\tau$$

$$= \frac{-10e^{-t}}{-2}\left[e^{-2\tau}\right]_0^t + \frac{20e^{-2t}}{(-1)}\left[e^{-\tau}\right]_0^t$$

$$= 5e^{-t}\,[e^{-2t} - 1] - 20\,e^{-2t}\,[e^{-t} - 1]$$

$$= (5e^{-3t} - 5e^{-t} - 20\,e^{-3t} + 20\,e^{-2t})\,u(t)$$

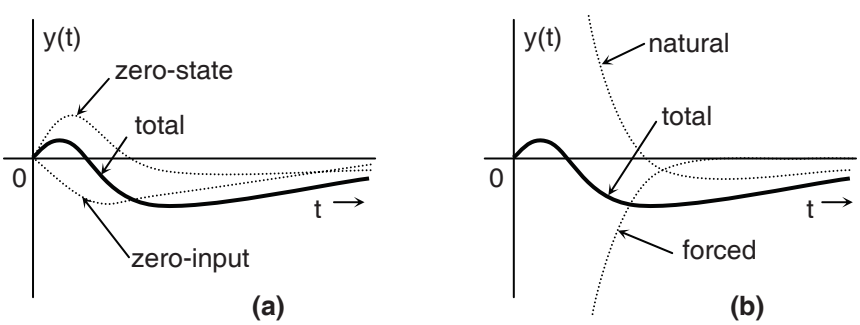
$y_{zs} = (-5e^{-t} + 20e^{-2t} - 15e^{-3t})_{u(t)}$ zero-state response

$\therefore$ Total response y(t) = zero-input response + zero-state response

$$= y_0(t) + y_{zs}(t)$$

$$= \underbrace{(-5e^{-t} + 5e^{-2t})\,u(t)}_{\text{zero-input response}} + \underbrace{(-5e^{-t} + 20e^{-2t} - 15e^{-3t})\,u(t)}_{\text{zero-state response}} \qquad \ldots(32)$$

Following figure (a) shows the zero-input, the zero-state, and the total response.



Total response and its components

Natural and Forced Response

From above example we see that the zero-input response is composed exclusively of characteristic modes. Also the zero-state response equation (32) contains characteristic mode terms.

We can now lump together all the characteristic mode terms in the total response, giving us a component known as the *natural response* $y_n(t)$. The remainder, consisting entirely of non-characteristic mode terms, is known as the *forced response* $y_\phi(t)$. The total response can be expressed in terms of natural and forced components

$$\text{Total response} = \underbrace{(-10e^{-t} + 25e^{-2t})}_{\text{natural response } y_n(t)} + \underbrace{(-15e^{-3t})}_{\text{forced response } y_\phi(t)} \qquad t \geq 0 \qquad \ldots(33)$$

Above figure (b) shows the natural, forced, and total response.

Classical Solution of Differential Equations

In the classical method we solve differential equation to find the natural and forced components rather than the zero-input and zero-state components of the response.

Note : Natural response $y_n(t)$ (also known as the *homogeneous solution* or *complementary solution*). The remaining portion of the response consists entirely of non-characteristic mode terms and is called the system's forced response $y_\phi(t)$ (also known as the *particular solution*.)

The natural response, being a linear combination of the system's characteristic modes, has the same form as that of the zero-input response; only its arbitrary constants are different and forced response $y_\phi(t)$ can be obtained using method of undetermining coefficients.

Forced Response: The Method of Undetermined Coefficients

	Input f(t)	Forced Response, $y_\phi(t)$
1.	$e^{\zeta t}, \ \zeta \neq \lambda_i (i = 1, 2, \cdots, n)$	$\beta e^{\zeta t}$
2.	$e^{\zeta t}, \ \zeta = \lambda_i$	$\beta t e^{\zeta t}$
3.	k	β
4.	$\cos(\omega t + \theta)$	$\beta \cos(\omega t + \phi)$
5.	$(t^r + \alpha_{r-1} t^{r-1} + \cdots + \alpha_1 t + \alpha_0) e^{\zeta t}$	$(\beta_r t^r + \beta_{r-1} t^{r-1} + \cdots + \beta_1 t + \beta_0) e^{\zeta t}$

Note :

By definition, $y_\phi(t)$ cannot have any characteristic mode terms. If any term appearing in the right-hand column for the forced response is also a characteristic mode of the system, the correct form of the forced response must be modified to $t^i y_\phi(t)$, where *i* is the smallest possible integer that can be used and still can prevent $t^i y_\phi(t)$ from having a characteristic mode term. For example, when the input is $e^{\zeta t}$, the forced response (right-hand column) has the form $\beta e^{\zeta t}$. But if $e^{\zeta t}$ happens to be a characteristic mode of the system, the correct form of the forced response is $\beta t e^{\zeta t}$ (see pair 2). If $te^{\zeta t}$ also happens to be a characteristic mode of the system, the correct form of the forced response is $\beta t^2 e^{\zeta t}$, and so on.

This method can be used only for inputs having with a finite number of derivatives; this class of inputs includes a wide variety of the most commonly encountered signals in practice.

Example:

Find total response (solving same example as above) : natural and forced response
$(D^2 + 3D + 2)\, y(t) = Df(t)$
$f(t) = 10e^{-3t}$ and $y(0^+) = 0$ and $\dot{y}(0^+) = 5$

Solution:

Note : here initial condution are given at $t = 0^+$, which applies to the total solution)

i) natural response $y_n(t)$:
The characteristic equation is $\lambda^2 + 3\lambda + 2 = 0$
$\Rightarrow$ $(\lambda + 2)(\lambda + 1) = 0$
$\Rightarrow$ $\lambda_1 = -1$ and $\lambda_2 = -2$
$\therefore$ $y_n(t) = k_1 e^{-t} + k_2 e^{-2t}$ …(34)

ii) Forced response, $y_\phi(t)$ (using method of undetermined co-efficient)

$\because$ $f(t) = 10e^{-3t}$ …[is not of the form of characteristic response]

let $y_\phi(t) = \beta e^{-3t}$
Substituting this in given D.E. we get

$$\ddot{y}_\phi(t) + 3\dot{y}_\phi(t) + 2y_\phi(t) = f(t)$$

$\Rightarrow$ $9\,\beta e^{-3t} + 3[-3\beta e^{-3t}] + 2\beta e^{-3t} = -30e^{-3t}$
$\Rightarrow$ $2\,\beta e^{-3t} = -30e^{-3t}$
comparing, we get $\beta = -15$
$\therefore$ $y_\phi(t) = -15\, e^{-3t}$ …(35)
= natural response + forced response
$y(t) = y_n(t) + y_\phi(t)$ …[from 34,35]
$y(t) = (k_1 e^{-t} + k_2 e^{-2t} - 15e^{-3t})u(t)$

To find k_1 and k_2, now, we apply initial conditions

$y(0^+) = 0,\ \dot{y}(0^+) = 5$

$\Rightarrow$ $k_1 + k_2 = 15$
$-k_1 - 2k_2 = -40$ …solving we get $k_1 = -10$, $k_2 = 25$
$y(t) = (-10e^{-t} + 25e^{-2t} - 15e^{-3t})u(t)$ …required total response

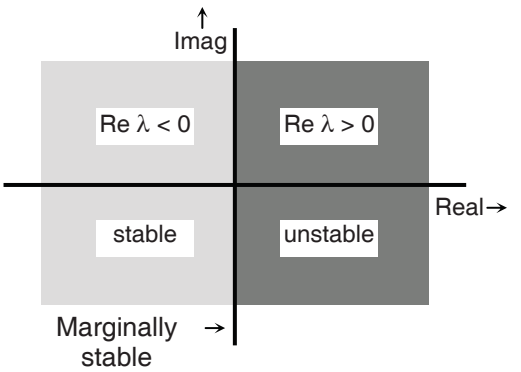
which is same as before (refer equation (33))

- advantage of using this method very simple
- disadvantage
  - can not be broken into zero-input and zero-state response
  - initial condition required are at $t = 0^+$, but generally we know condition at $t = 0^-$, so these conditions must be derived

Note : In the classical method, the initial conditions are required at $t = 0^+$. The reason is that because at $t = 0^-$, only the zero-input component exists, and the initial conditions at $t = 0^-$ can be applied to the zero-input component only. In the classical method, the zero-input and zero-state components cannot be separated. Consequently, the initial conditions must be applied to the total response, which begins at $t = 0^+$.

SYSTEM STABILITY

If, in the absence of an external input, a system remains in a particular state (or condition) indefinitely, then that state is said to be an equilibrium state of the system (zero state). Now suppose an LTI system is in equilibrium (zero state) and we change this state by creating some nonzero initial conditions. If the system is stable it should eventually return to zero state. But the system output generated by initial condtions (zero-input response) is made up of its characteristic modes. For this reason we define stability as follows: a system is *(asymptotically) stable* if, and only if, all its characteristic modes $\rightarrow 0$ as $t \rightarrow \infty$. If any of the modes grows without bound as $t \rightarrow \infty$, the system is unstable. If zero-input response remains bounded (approaches neither zero nor infinity), approaching a constant or oscillating with a constant amplitude as $t \rightarrow \infty$.



Characteristic roots location and system stability

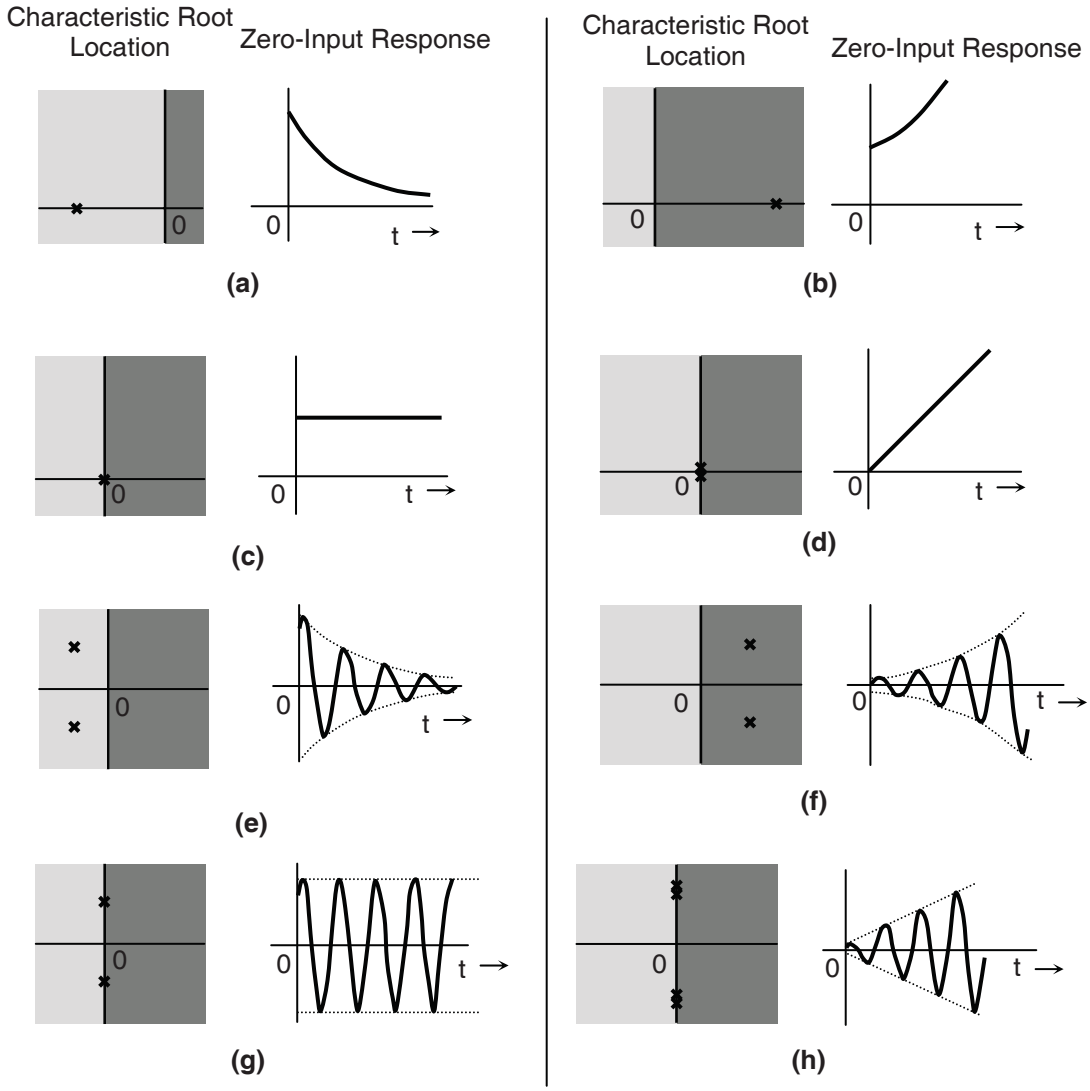
For this borderline situation, the system is said to be marginally stable or just stable.

To summarize :

1. An LTIC system is asymptotically stable if, and only if, all the characteristic roots are in the LHP. The roots may be simple (unrepeated) or repeated.

2. An LTIC system is unstable if, and only if, either one or both of the following conditions exist:
   (i) at least one root is in the RHP, (ii) there are repeated roots on the imaginary axis.

3. An LTIC system is marginally stable if, and only if, there are no roots in the RHP, and there are some unrepeated roots on the imaginary axis.



Location of characteristic roots and the corresponding characteristic modes

Alternative form for System Stability

Consequently, for an asymptotically stable system

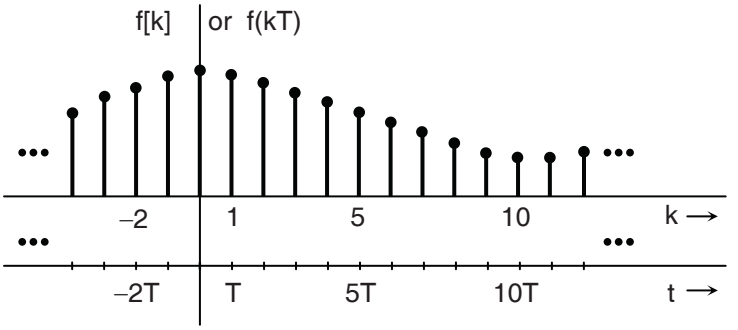
$$\int_{-\infty}^{\infty} | h(\tau) | \, d\tau < K_2 < \infty$$

These result lead to the formulation of an alternative definition of stability known as bounded-input, bounded-output (BIBO) stability: a system is BIBO stable if, and only if, a bounded input produces a bounded output. Observe that an asymptotically stable system is always BIBO stable. However, a marginally stable system is BIBO unstable.

PART(B) : DISCRETE TIME SIGNALS AND SYSTEM

Introduction

A typical discrete-time signal is a sequence of numbers specified at uniform interval of time. This signal may be viewed as a function of time t where signal values are specified at t = kT or may be viewed as a function of k (k, integer) as shown in the following figure.



A discrete-time signal

The representation f[k] is more convenient and will be followed throughout this book.

Advantages of Digital Signal Processing

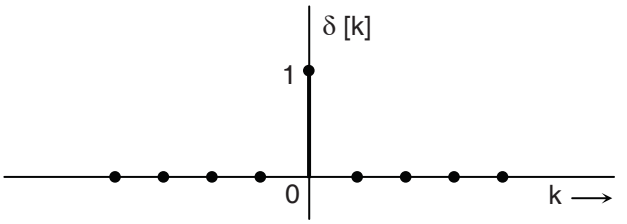
1. Digital filters have a greater degree of precision and stability.
2. Digital filters are more flexible.
3. A greater variety of filters can be realized by digital systems.
4. Very low frequency filters, if realized by continuous-time systems, require prohibitively bulky components. Such is not the case with digital filters.
5. Digital signals can be stored easily on magnetic tapes or disks without deterioration of signal quality.
6. More sophisticated signal processing algorithms can be used to process digital signals.
7. Digital filters can be time shared, and therefore can serve a number of inputs simultaneously.
8. Using integrated circuit technology, they can be fabricated in small packages requiring low power consumption.

TYPES OF SIGNALS

1. **Discrete-Time Impulse Function, δ[k]**
   The discrete-time counterpart of the continuous-time impulse function δ(t) is δ[k], defined by

$$\delta[k] = \begin{cases} 1 & k = 0 \\ 0 & k \neq 0 \end{cases}$$

This function, also called the unit impulse sequence, is shown in following figure. Unlike its continuous-time counterpart $\delta(t)$, this is a very simple function without any mystery.



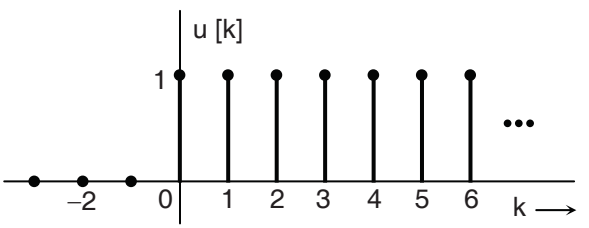
Discrete-time impulse function

2. Discrete-Time Unit Step Function, u[k]

The discrete-time counterpart of the unit step function u(t) is u[k] defined by

$$u[k] = \begin{cases} 1 & \text{for } k \geq 0 \\ 0 & \text{for } k < 0 \end{cases}$$

If we want a signal to start at k = 0 (so that it has a zero value for all k < 0), we need only multiply the signal with u[k]. The following figure shows u[k]



A discrete-time unit step function u[k]

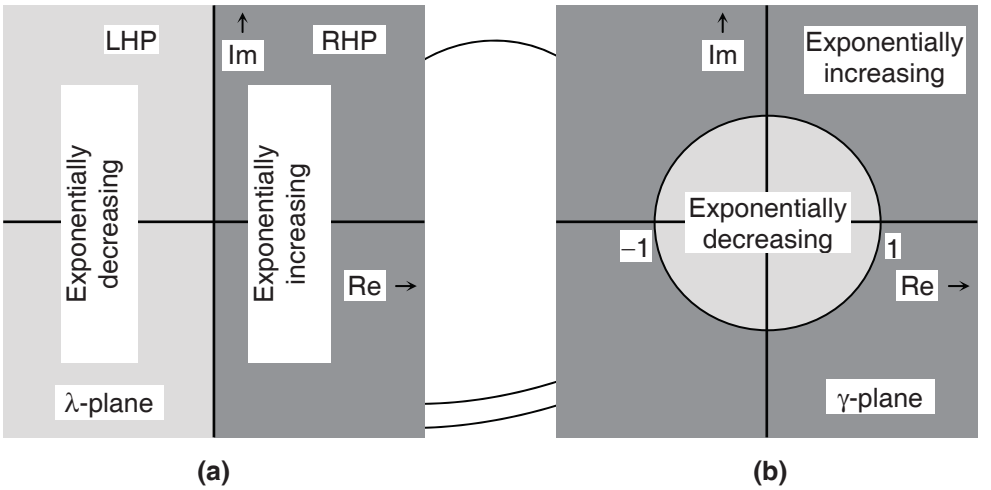
3. Discrete-Time Exponential, γ^k

The discrete-time exponential can be expressed in two forms as

$$e^{\lambda k} = \gamma^k \qquad (\gamma = e^{\lambda} \text{ or } \lambda = \ln \gamma)$$

In the study of discrete-time signals and systems, unlike the continuous-time case, the form γ^k proves more convenient than the form $e^{\lambda k}$.

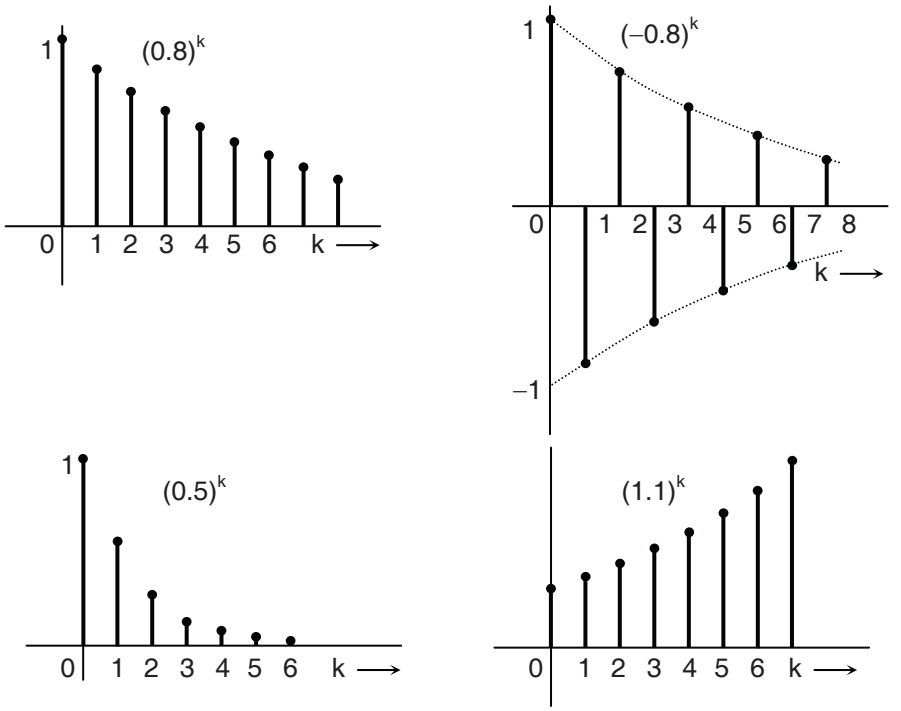
Nature of γ^k : The signal $e^{\lambda k}$ grows exponentially with k if Re $\lambda > 0$ (λ in RHP), and decays exponentially if Re $\lambda < 0$ (λ in LHP). It is constant or oscillates with constant amplitude if Re $\lambda = 0$ (λ on the imaginary axis). Clearly, the location of λ in the complex plane indicates whether the signal $e^{\lambda k}$ grows exponentially, decays exponentially, or oscillates with constant frequency [refer following figure (a)]. A constant signal ($\lambda = 0$) is also an oscillation with zero frequency. A similar criterion for determining the nature of γ^k from the location of γ in the complex plane [refer following figure (b)].



The λ-plane, the γ-plane and their mapping

- The imaginary axis in the λ-plane maps into the unit circle in the γ-plane.
- The left-half plane in the λ-plane maps into the inside of the unit circle.
- The right-half of the λ-plane maps into the outside of the unit circle in the γ-plane, as depicted in above figure (b).

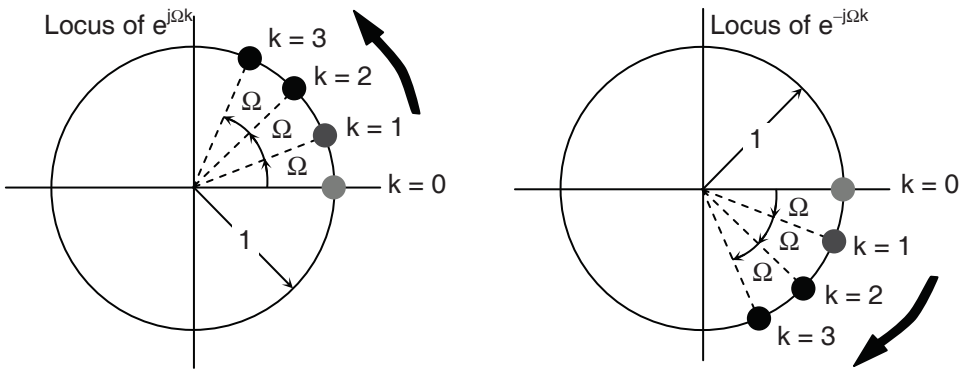
This fact means that the signal γ^k grows exponentially with k if γ is outside the unit circle ($|\gamma| > 1$), and decays exponentially if γ is inside the unit circle ($|\gamma| < 1$).The signal is constant or oscillates with constant amplitude if γ is on the unit circle ($|\gamma| = 1$).



Discrete-time exponentials γ^k

4. Discrete-Time Exponential, $e^{j\Omega k}$

A general discrete-time exponential $e^{j\Omega k}$ (also called phasor) is a complex valued function of k, we shall plot the values of $e^{j\Omega k}$ in the complex plane for various values of k, as illustrated in following figure. The function f[k] = $e^{j\Omega k}$ takes on values e^{j0}, $e^{j\Omega}$, $e^{j2\Omega}$, $e^{j3\Omega}$, … at k = 0, 1, 2, 3, … , respectively.



(a) **(b)**

Note that

$$e^{j\Omega k} = re^{j\theta}, \quad r = 1, \quad \text{and} \quad \theta = k\Omega$$

The locus of $e^{j\Omega k}$ may be viewed as a phasor rotating counterclockwise at a uniform speed of Ω radians per unit sample interval. The exponential $e^{-j\Omega k}$, on the other hand, takes on values $e^{j\theta} = 1$, $e^{-j\Omega}$, $e^{-j2\Omega k}$, $e^{-j3\Omega}$, … at k = 0, 1, 2, 3, …, as depicted in above figure (b). Therefore, $e^{-j\Omega k}$ may be viewed as a phasor rotating clockwise at a uniform speed of Ω radians per unit sample interval.

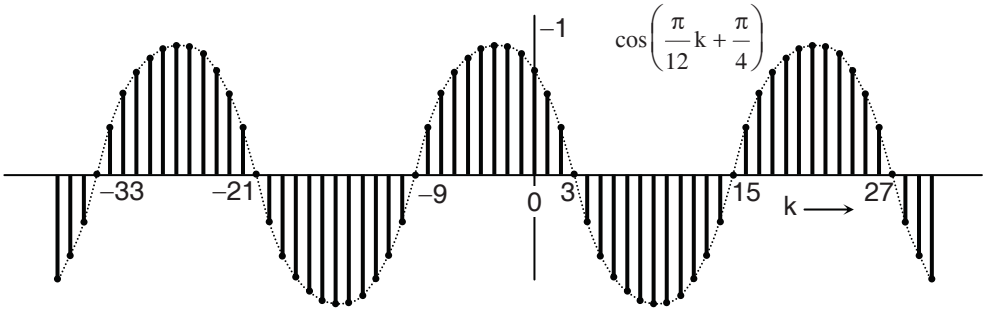
Using Euler's formula, we can express an exponential $e^{j\Omega k}$ in terms of sinusoids of the form cos ($\Omega k + \theta$), and vice versa.

$$e^{j\Omega k} = (\cos \Omega k + j \sin \Omega k)$$

$$e^{-j\Omega k} = (\cos \Omega k - j \sin \Omega k)$$

5. Discrete-Time sinusoid, cos($\Omega k + \theta$)

A general discrete-time sinusoid can be expressed as C cos ($\Omega k + \theta$), where C is the amplitude, Ω is the frequency (in radians per sample), and θ is the phase (in radians). Following figure shows a discrete-time sinusoid $\cos\left(\frac{\pi}{12}k + \frac{\pi}{4}\right)$.



A discrete-time sinusoid $\cos\left(\frac{\pi}{12}k+\frac{\pi}{4}\right)$

- **Some Peculiarities of Discrete-Time Sinusoids**

  There are three unexpected properties of discrete-time sinusoids which distinguish them from their continuous-time relatives.

  a. In the continuous-time case, the period of a sinusoid can take on any value; integral, fractional, or even irrational. The discrete-time signal, in contrast, is specified only at integral values of k. Therefore, the period must be an integer (in terms of k) or an integral multiple of T (in terms of variable t).

  b. A continuous-time sinusoid is always periodic regardless of the value of its frequency ω. But a discrete-time sinusoid cos Ωk is periodic only if Ω is 2π times some rational number $\left(\frac{\Omega}{2\pi} \text{ is a rational number}\right)$.

  c. A continuous-time sinusoid cos ωt has a unique waveform for each value of ω. In contrast, a sinusoid cos Ωk does not have a unique waveform for each value of Ω. In fact, discrete-time sinusoids with frequencies separated by multiples of 2π are identical. Thus, a sinusoid cos Ωk = cos $(\Omega + 2\pi)$ k = cos $(\Omega + 4\pi)$ k = …

- **Not All Discrete-Time Sinusoids Are Periodic**

  A discrete-time signal f[k] is said to be N_0-periodic if

  $$f[k] = f[k + N_0] \qquad \text{…(36)}$$

  for some positive integer N_0. The smallest value of N_0 that satisfies equation (36) is the period of f[k].

  If a signal cos Ωk is to be N_0-periodic, then

  $$\text{Cos } \Omega k = \cos \Omega (k + N_0) = \cos (\Omega k + \Omega N_0)$$

  This result is possible only if ΩN_0 is an integral multiple of 2π; that is,

  $$\Omega N_0 = 2\pi m \quad m \text{ integer}$$

  $$\frac{\Omega}{2\pi} = \frac{m}{N_0} \qquad \text{…(37)}$$

  Because both m and N_0 are integers, equation (37) implies that the sinusoid cos Ωk is periodic only if $\frac{\Omega}{2\pi}$ is a rational number.

Example :

State with reasons if the following sinusoids are periodic. If periodic, find the period.

(i) $\cos\left(\frac{3\pi}{7}k\right)$ (ii) $\cos(\sqrt{\pi}k)$

Solution:

(i) $\cos\left(\frac{3\pi}{7}k\right)$ comparing with standard form $\cos \Omega k$, we get $\Omega = \frac{3\pi}{7}$

$$\frac{\Omega}{2\pi} = \frac{3\pi}{7}\cdot\frac{1}{2\pi} = \frac{3}{14} \quad \left(\text{which is of the form } \frac{m}{N_0}\right)$$

Thus, this sinusoid is periodic with N_0 (period) equal to 14.

(ii) $\cos\left(\sqrt{\pi}k\right)$

Similarly, here $\Omega = \sqrt{\pi}$

$$\therefore \frac{\Omega}{2\pi} = \frac{\sqrt{\pi}}{2(\sqrt{\pi})^2} = \frac{1}{2\sqrt{\pi}} \text{ (irrationational)}$$

$\therefore$ The given sinusoid is aperiodic

- **Non-uniqueness of Discrete-Time Sinusoid Waveforms**

A continuous-time sinusoid $\cos \omega t$ has a unique waveform for every value of ω in the range 0 to ∞. Increasing ω results in a sinusoid of ever increasing frequency. Such is not the case for the discrete-time sinusoid $\cos \Omega k$ because

$$\cos(\Omega \pm 2\pi m)\, k = \cos(\Omega k \pm 2\pi m k)$$

Now, if m is an integer, mk is also an integer, and the above equation reduces to

$$\cos(\Omega \pm 2\pi m)\, k = \cos \Omega k, \quad m \text{ integer}$$

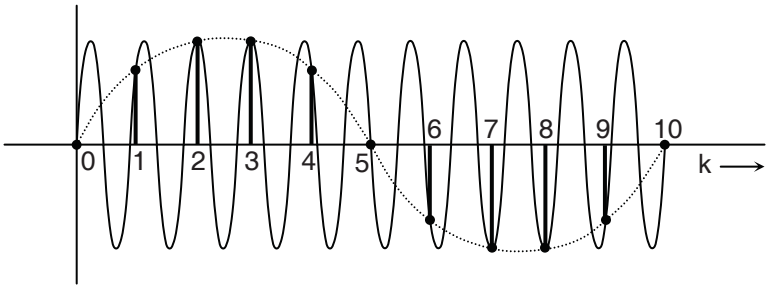
This result shows that a discrete-time sinusoid of frequency Ω is indistinguishable from a sinusoid of frequency Ω plus or minus an integral multiple of 2π. This statement certainly does not apply to continuous-time sinusoids.

This result means that discrete-time sinusoids of frequencies separated by integral multiples of 2π are identical. Thus the fundamental range of frequencies is from $-\pi$ to π.

Thus $\cos(8.7\pi k + \theta) = \cos(0.7\pi k + \theta)$

$\cos(9.6\pi k + \theta) = \cos(-0.4\pi k + \theta)$

This result shows that a sinusoid $\cos(\Omega k + \theta)$ can always be expressed as $\cos(\Omega_f k + \theta)$, where $-\pi \le \Omega_f < \pi$ (the fundamental frequency range). The graphical explanation is shown in the figure below, which shows that the given discrete time sinusoid may be sample of two continuous time sinusoids.



Physical explanation of non-uniqueness of Discrete-time sinusoid waveforms

Further Reduction in the Frequency Range of Distinguishable Discrete-Time Sinusoids

The frequencies in the range (0 to $-\pi$) can be expressed as frequencies in the range (0 to π) with opposite phase.

$$\cos(9.6\pi k + \theta) = \cos(-0.4\pi k + \theta) = \cos(0.4\pi k - \theta)$$

This result shows that a sinusoid of any frequency Ω can always be expressed as a sinusoid of a frequency $|\Omega_f|$, where $|\Omega_f|$ lies in the range 0 to π.

Example :

Consider sinusoids of frequencies Ω equal to (a) 1.6 π (b) 2.5 π. Each of these sinusoids is equivalent to a sinusoid of some frequency $|\Omega_f|$ in the range 0 to π. Determine $|\Omega_f|$.

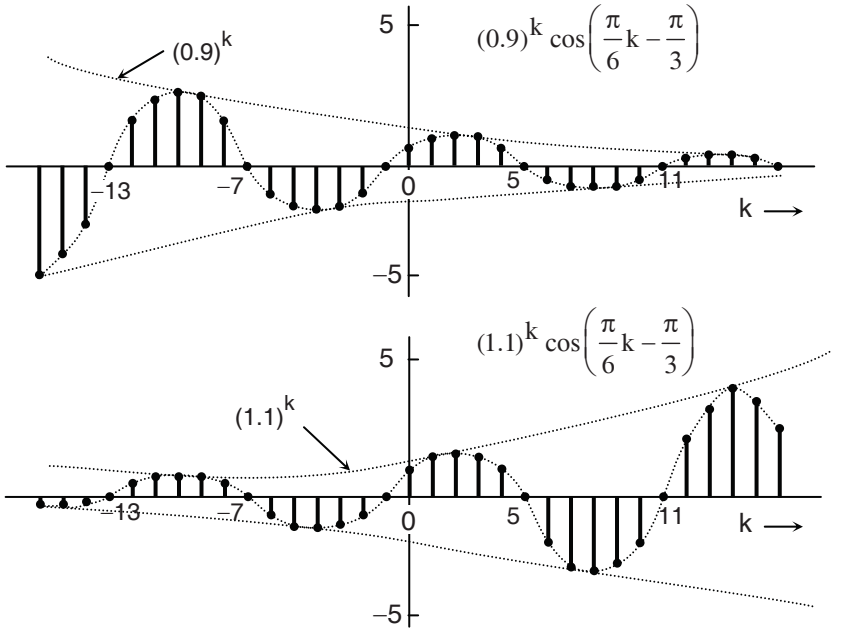
Solution :

(a) The frequency $1.6\pi = 2\pi - 0.4\pi$, and $\Omega_f = -0.4\pi$. Therefore, a sinusoid of frequency 1.6π can be expressed as a sinusoid of frequency $|\Omega_f| = 0.4\pi$.

(b) $2.5\pi = 2\pi + 0.5\pi$, and $\Omega_f = 0.5\pi$. Therefore, a sinusoid of frequency 2.5π can be expressed as a sinusoid of frequency $|\Omega_f| = 0.5\pi$.

6. Exponentially Varying Discrete-Time Sinusoid, $\gamma^k \cos(\Omega k + \theta)$

This is a sinusoid $\cos(\Omega k + \theta)$ with an exponentially varying amplitude γ^k. It is obtained by multiplying the sinusoid $\cos(\Omega k + \theta)$ by an exponential γ^k. Following figure shows $(0.9)^k \cos\left(\frac{\pi}{6}k - \frac{\pi}{3}\right)$, and $(1.1)^k \cos\left(\frac{\pi}{6}k - \frac{\pi}{3}\right)$. Observe that if $|\gamma| < 1$, the amplitude decays, and if $|\gamma| > 1$, the amplitude grows exponentially.



Examples of Exponentially Varying Discrete-Time Sinusoids

Energy And Power For Discrete Time Signals

- *Energy of a signal*

  The energy of the discrete time signals is given by

  $$E = \sum_{k=-\infty}^{\infty} |f[k]|^2$$

  For the signal to be energy signal, $E < \infty$ (i.e. it must be finite). Thus for energy signal $f[k] \to 0$ as $k \to \infty$

- *Power of a signal*

  The power of discrete time signal is given by

  $$P_{avg} = \lim_{k\to\infty} \frac{1}{2N+1} \sum_{k=-N}^{N} |f[k]|^2$$

  For the signal to be power signal, $P_{avg} < \infty$. (i.e. it must be finite)

  Example :

  1. $f[k] = a^k u[k]$

     $$E = \sum_{k=-\infty}^{\infty} |f[k]|^2 = \sum_{k=0}^{\infty} a^{2k} = \sum_{k=0}^{\infty} (a^2)^k$$

     $$= \frac{1}{1-a^2} \quad \to \text{Energy signal}$$

2. f[k] = u[k]

$$E = \sum_{k=-\infty}^{\infty} |f[k]|^2 = \sum_{k=0}^{\infty} |u[k]|^2 = \infty \rightarrow \text{Power signal}$$

or we define $E_N = \sum_{k=0}^{N} |f[k]|^2 = \sum_{k=0}^{N} |f[k]|^2$

$$= N + 1$$

then $E = \underset{N\to\infty}{\text{Lt}}\, E_N = \infty$

To find power $= \underset{N\to\infty}{\text{Lt}} \frac{1}{2N+1} \sum_{k=0}^{N} |f[k]|^2$

$$P = \underset{N\to\infty}{\text{Lt}} \frac{1}{2N+1} \sum_{k=0}^{N} u^2 = \underset{N\to\infty}{\text{Lt}} \frac{1}{2N+1} \sum_{k=0}^{N} 1^2$$

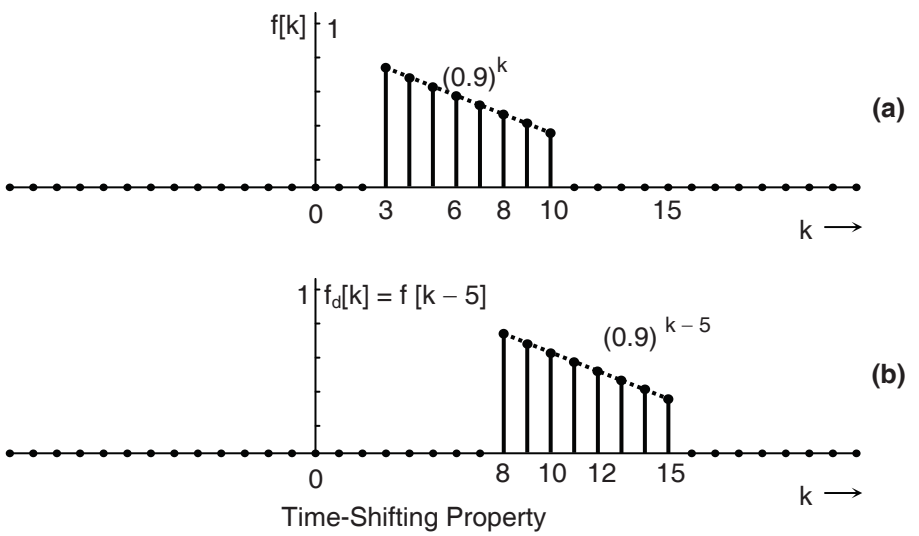
$$= \underset{N\to\infty}{\text{Lt}} \frac{N+1}{2N+1} \quad \underset{N\to\infty}{\text{Lt}} \frac{N+1}{2N+1} = \frac{1}{2}$$

USEFUL SIGNAL OPERATIONS

Signal operations discussed for continuous-time systems also apply to discrete-time systems with some modification in time scaling.

(A) Time Shifting

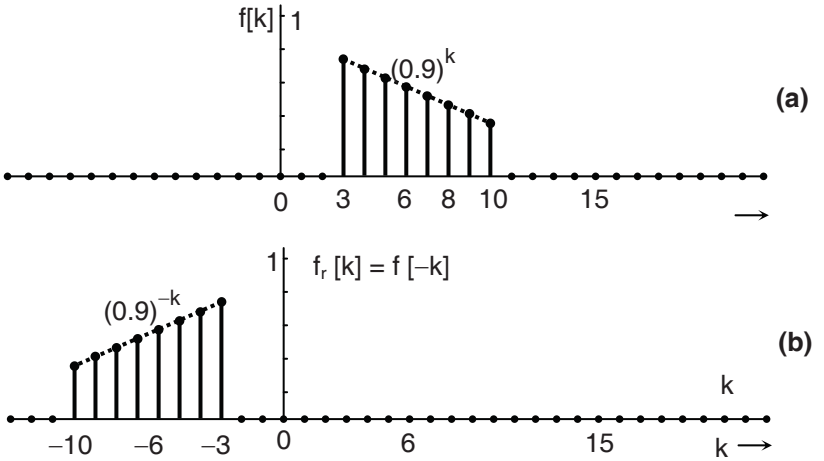
A signal f[k] may be shifted in time by replacing the independent variable k by k–m, where m is an integer. If m is a positive integer, the time shift results in a delay of the signal by m units of time. If m is a negative integer, the time shift results in an advance of the signal by m units of time, as shown in following figure.



Time-Shifting Property

(B) Time Inversion (or Reversal)

We can show that to time invert a signal f[k], we replace k with –k. This operation rotates the signal about the vertical axis, as shown in following figure.



Time inversion of a signal

(C) Time Scaling

Time Compression: Decimation or Downsampling

Consider a signal

$$f_c[k] = f[2k]$$

The signal $f_c[k]$ is the signal f[k] compressed by a factor 2. Observe that $f_c[0] = f[0]$, $f_c[1] = f[2]$, $f_c[2] = f[4]$, and so on. This fact shows that $f_c[k]$ is made up of even numbered samples of f[k]. The odd numbered samples of f[k] are missing as shown in following figure (b). This operation loses part of the data, and that is why such time compression is called *decimation* or *downsampling.* In the continuous-time case, time compression merely speeds up the signal without loss of data. In general, f[mk] (m integer) consists of only every m th sample of f[k].

Time Expansion

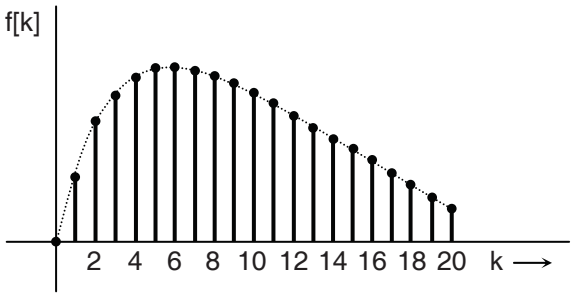
Consider a signal

$$f_e[k] = f\left[\frac{k}{2}\right] \qquad \ldots(38)$$

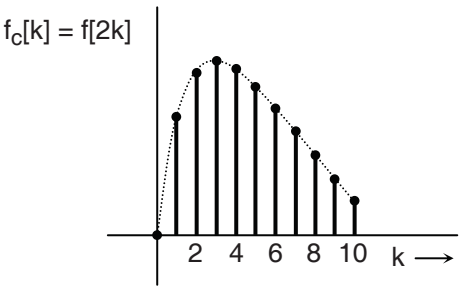
The signal $f_e[k]$ is the signal f[k] expanded by a factor 2. According to equation (38), $f_e[0] = f[0]$, $f_e[1] = f[1/2]$, $f_e[2] = f[1]$, $f_e[3] = f[3/2]$, $f_e[4] = f[2]$, $f_e[5] = f[5/2]$, $f_e[6] = f[3]$, and so on. Now, f[k] is defined only for integral values of k, and is zero (or undefined) for all fractional values of k. Therefore, for $f_e[k]$, its odd numbered samples $f_e[1]$, $f_e[3]$, $f_e[5]$, …are all zero (or undefined), as depicted in following figure (c).

Interpolation

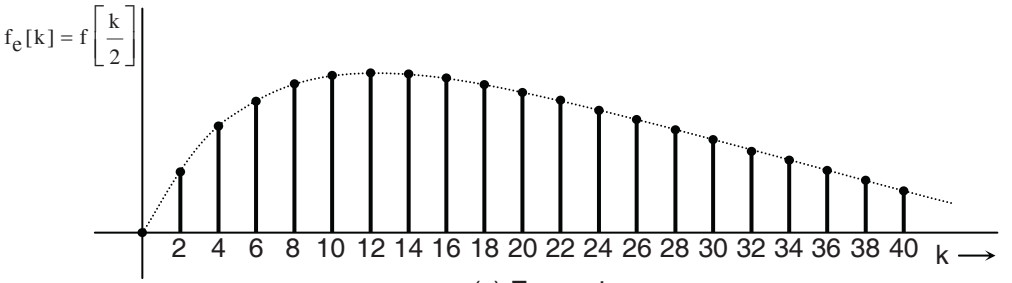
In the time-expanded signal in following figure (c), the missing odd numbered samples can be reconstructed from the non-zero valued samples using some suitable interpolation formula. We may use a realizable interpolation, such as a linear interpolation, where $f_i[1]$ is taken as the mean of $f_i[0]$ and $f_i[2]$. Similarly, $f_i[3]$ is taken as the mean of $f_i[2]$ and $f_i[4]$, and so on. This process of time expansion and inserting the missing samples using an interpolation is called *interpolation or upsampling.*



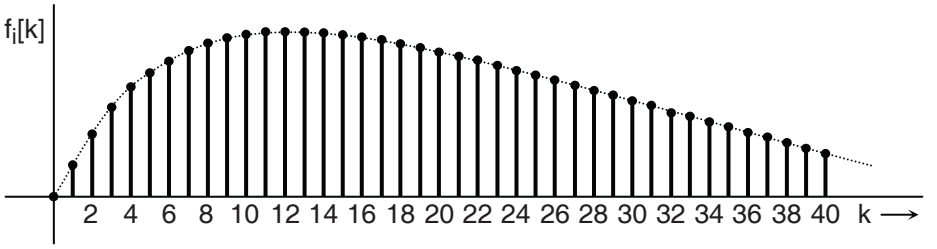
(a) Original Signal



(b) Decimation (Downsampling)



(c) Expansion



(d) Interpolation (Upsampling)

CLASSIFICATION OF SYSTEMS

(A) Static Vs Dynamic Systems

A discrete time system is called static or memory less if its output at any instant depends on the present input and not on past or future values of input. In any other case, the system is said to be dynamic or to have memory.

e.g. $y[k] = a\, f[k]$ static

$y[k] = \sum_{m=0}^{k} f[k-m]$ dynamic

(B) Time Invariant Vs Time Variant Systems

A system is said to be time invariant if the input – output characteristics do not change with time. If they change with time, then the system is time variant.

Example:

i) $y[k] = f[k] - f[k-1]$

$\therefore\ y[k-m] = f[k-m] - f[k-m-1] \rightarrow$ Time invariant

ii) $y[k] = k\, f[k]$

$\therefore\ y[k-m] = (k-m)\, f[k-m] = k\, f[k-m] - m\, f[k-m] \rightarrow$ Time variant

(C) Linear Vs Non Linear System

A linear system is one that satisfies *superposition principle*. A system H is linear if and only if

$$H\,[a_1 f_1[k] + a_2 f_2[k]] = a_1 H[f_1[k]] + a_2 H[f_2[k]]$$

for any arbitrary sequence $f_1[k]$ and $f_2[k]$ and any arbitrary constants a_1 and a_2. A relaxed system that does not satisfy the principle of superposition is called nonlinear. If $y[k] \neq 0$ for the input $f[k] = 0$, then the system is nonlinear [A linear system with 0 input produces 0 output].

$y[k] = f[k^2] \rightarrow$ linear

$y[k] = f^2[k] \rightarrow$ non linear

(D) Causal Vs Non Causal System

A system is said to be causal if the output of the system at any time k depends only on present and past inputs but does not depend on future inputs. The output of a causal system satisfies the equation $y[k] = f(f[k], f[k-1], f[k-2] \ldots\ldots)$. If the system does not satisfy this condition, it is called non causal

Example :

$y[k] = f[k] - f[k-1]$ causal
$y[k] = f[k^2]$ non causal

> Following are the conditions for Causal System :
> - Signal is right handed
> - Output does not depends on further input.

(E) Stable Vs Unstable System

An arbitrary relaxed system is said to be bounded input–bounded output (stable) if and only if every bounded input produces a bounded output. In other words,

Impulse response should be absolutely summable.
i.e. $\Sigma \mid h[k] \mid < \infty$
i.e. $\Sigma \mid h[k] \mid$ should be finite

(F) Invertible Vs Non Invertible System

A system is said to be invertible if there is a one correspondence between its input and output signals. i.e. if we know the output sequence $y[k]$ $-\infty \le k \le \infty$ of an invertible system, we can find its input $f[k]$ $-\infty \le k \le \infty$.

Example:

$y[k] = a\, f[k] \rightarrow$ invertible
$y[k] = f^2[k] \rightarrow$ non invertible

TIME-DOMAIN ANALYSIS OF LTID SYSTEM

Introduction

The procedure for analysis is parallel to that for continuous-time systems, with minor differences. A general nth-order difference equation, using advance operator form:

$$y[k+n] + a_{n-1y}[k+n-1] + \cdots + a_1 y[k+1] + a_0 y[k]$$

$$= b_m f[k+m] + b_{m-1} f[k+m-1] + \cdots + b_1 f[k+1] + b_0 f[k] \qquad \ldots(39)$$

Causality Condition

The left-hand side of equation (39) consists of the output at instants k + n, k + n – 1, k + n – 2, and so on. The right-hand side equation (39) consists of the input at instants k + m, k + m – 1, k + m – 2, and so on. For a causal system the output cannot depend on future input values. The causality requires m ≤ n. Therefore, we replace k by k – n throughout the equation yields the alternative form (the delay operator form) of equation (39) for m = n

$$y[k] + a_{n-1y}[k-1] + \cdots + a_1 y[k-n+1] + a_0 y[k-n]$$
$$= b_n f[k] + b_{n-1} f[k-1] + \cdots + b_1 f[k-n+1] + b_0 f[k-n] \qquad \ldots(40)$$

We shall designate form (39) the *advance operator form*, and form (40) the *delay operator form*.

Initial Conditions and Iterative Solution of Difference Equations

Equation (40) can be expressed as

$$y[k] = -a_{n-1y}[k-1] - a_{n-2y}[k-2] - \cdots - a_{0y}[k-n] + b_n f[k] + b_{n-1} f[k-1] + \cdots + b_0 f[k-n]$$

This equation shows that y[k], the output at the kth instant, is computed from 2n + 1 pieces of information. These are the past *n* values of the output: y[k – 1], y[k – 2] ,…, y[k – n], the past n values of the input: f [k – 1], f [k – 2],…, f [k – n], and the present value of the input f[k]. If the input is causal, then f[–1] = f[–2] = … = f[–n] = 0, and we need only n initial conditions y[–1], y[–2],…, y[–n]. This result allows us to compute iteratively or recursively the output y[0], y[1], y[2], y[3], … , and so on.

Example

Solve iteratively

$$y[k] - 0.5y[k-1] = f[k]$$

with initial condition y[–1] = 16 and causal input f[k] = k^2 (starting at k = 0).

Solution :

This equation can be expressed as

$$y[k] = 0.5y[k-1] + f[k] \qquad \ldots(41)$$

If we set k = 0 in this equation, we obtain

$$y[0] = 0.5y[-1] + f[0]$$
$$= 0.5(16) + 0 = 8$$

Now, setting k = 1 in equation (41) and using the value y[0] = 8 (computed in the first step) and

f[1] = $(1)^2$ = 1, we obtain

y[1] = 0.5(8) + $(1)^2$ = 5

Next, setting k = 2 in equation (41) and using the value y[1] = 5

(computed in the previous step)

and $f[2] = (2)^2$, we obtain

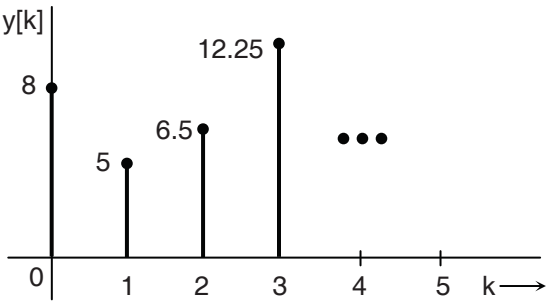
$$y[2] = 0.5(5) + (2)^2 = 6.5$$

Continuing in this way iteratively, we obtain

$$y[3] = 0.5(6.5) + (3)^2 = 12.25$$

$$y[4] = 0.5(12.25) + (4)^2 = 22.125$$

The output y[k] is depicted in figure below.



Iterative solution of a difference equation

Iterative method can be applied to a difference equation in delay or advance operator form. The iterative procedure cannot be applied to continuous-time systems. It is interesting to note that the hardware realization generates the solution precisely in this (iterative) fashion.

Operational Notation

In difference equations it is convenient to use operational notation similar to that used in differential equations for the sake of compactness. For discrete-time systems we shall use the advance operator E to denote advancing of the sequence by one time unit. Thus

$$\begin{aligned} Ef[k] &\equiv f[k+1] \\ E^2 f[k] &\equiv f[k+2] \\ &\ldots\ldots\ldots\ldots \\ E^n f[k] &\equiv f[k+n] \end{aligned}$$

A general n^{th}–order difference equation (39), for m = n can be expressed as

$$(E^n + a_{n-1}E^{n-1} + \cdots + a_1 E + a_0)y[k] = (b_n E^n + b_{n-1}E^{n-1} + \cdots + b_1 E + b_0)f[k] \quad \ldots(42)$$

or $\quad Q[E]\, y[k] = P[E]\, f[k]$

where Q[E]and P[E] are nth-order polynomial operators

$$Q[E] = E^n + a_{n-1}E^{n-1} + \cdots + a_1 E + a_0$$

$$P[E] = b_n E^n + b_{n-1}E^{n-1} + \cdots + b_1 E + b_0$$

System response to Internal Conditions: The zero-Input response

The zero-input response $y_0[k]$ is the solution of equation (42) with f[k] = 0; that is,

$$Q[E]y_0[k] = 0 \qquad \ldots(43)$$

This equation states that a linear combination of $y_0[k]$ and advanced $y_0[k]$ is zero not for some values of k, but for all k. Such situation is possible if and only if $y_0[k]$ and advanced $y_0[k]$ have the same form. Only an exponential function γ^k has this property.

Therefore, the solution of equation (43) must be of the form

$$y_0[k] = c\gamma^k$$

Substitution of these yields

$$c(\gamma^n + a_{n-1}\gamma^{n-1} + \cdots + a_1\gamma + a_0)\gamma^k = 0$$

For a nontrivial solution of this equation

$$(\gamma^n + a_{n-1}\gamma^{n-1} + \cdots + a_{1\gamma} + a_0) = 0$$

$$Q[\gamma] = 0 \qquad \ldots(44)$$

Thus $c\gamma^k$ is indeed the solution provided γ satisfies equation (44)

The polynomial $Q[\gamma]$ is called the *characteristic polynomial* of the system, and $Q[\gamma] = 0$ is the characteristic equation of the system. Moreover, $\gamma_1, \gamma_2, \ldots\gamma_n$, the roots of the characteristic equation, are called *characteristic roots* or *characteristic values (also eigenvalues)* of the system. The exponentials $\gamma_i^k (i = 1, 2, \ldots, n)$ are the *characteristic modes* or *natural modes* of the system. Characteristic modes not only determine the zero-input response but also plays an important role in determining the zero-state response.

The zero input response, $y_0[k]$ depends upon the nature of the characteristic roots of equation $Q(\gamma) = 0$.

Nature of roots of $Q(\gamma) = 0$	Zero-input response, $y_0[k]$
1. All distinction roots	$y_0[k] = c_1e^{\gamma_1 k} + c_2e^{\gamma_2 k} + \cdots + c_ne^{\gamma_n k}$
2. Some roots repeated (say r-times)	$y_0[k] = (c_1 + c_2k + c_3k^2 + \cdots + c_rk^{r-1})e^{\gamma k}$
3. Complex conjugate roots of the form $\alpha \pm j\beta = \lvert\gamma\rvert e^{\pm j\beta}$	$y_0[k] = ce^{\gamma k}\cos(\beta k + \theta)$

Example

1. For an LTID system described by the difference equation

$$y[k+2] - 0.6y[k+1] - 0.16y[k] = 5f[k+2]$$

find zero-input response if the initial condition are $y[-1] = 0$ and $y[-2] = \frac{25}{4}$,

Solution:

The system equation in operational notation is

$$(E^2 - 0.6E - 0.16)\,y[k] = 5E^2 f[k]$$

The characteristic polynomial is

$$\gamma^2 - 0.6\gamma - 0.16 = (\gamma + 0.2)(\gamma - 0.8)$$

The characteristic equation is

$$(\gamma + 0.2)\ (\gamma - 0.8) = 0$$

The characteristic roots are $\gamma_1 = -0.2$ and $\gamma_2 = 0.8$. The zero-input response is

$$y_0[k] \quad = \quad c_1(-0.2)^k + c_2(0.8)^k \qquad \ldots(45)$$

To determine arbitrary constants c_1 and c_2, we set $k = -1$ and -2 in equation (45) then substitute

$y_0[-1] = 0$ and $y_0[-2] = \dfrac{25}{4}$ to obtain

$$\left.\begin{aligned} 0 &= -5c_1 + \frac{5}{4}c_2 \\ \frac{25}{4} &= 25c_1 + \frac{25}{16}c_2 \end{aligned}\right\} \quad \Rightarrow \quad \begin{aligned} c_1 &= \frac{1}{5} \\ c_2 &= \frac{4}{5} \end{aligned}$$

Therefore

$$y_0[k] = \frac{1}{5}(-0.2)^k + \frac{4}{5}(0.8)^k$$

2. A similar procedure may be followed for repeated roots. For instance, for a system specified by the equation

$$(E^2 + 6E + 9)\ y[k] = (2E^2 + 6E)\,f[k]$$

determine $y_0[k]$, the zero-input component of the response if the initial condition are

$$y_0[-1] = -\frac{1}{3} \qquad \text{and} \qquad y_0[-2] = -\frac{2}{9}.$$

Solution :

The characteristic polynomial is $\gamma^2 + 6\gamma + 9 = (\gamma + 3)^2$, and we have a repeated characteristic root at $\gamma = -3$. The characteristic modes are $(-3)^k$ and $k(-3)^k$.

Hence, the zero-input response is

$$y_0[k] = (c_1 + c_2 k)\ (-3)^k$$

We can determine the arbitrary constants c_1 and c_2 from the initial conditions following the procedure in example (1).

We find that $c_1 = 4$ and $c_2 = 3$ so that

$$y_0[k] = (4 + 3k)\ (-3)^k$$

3. For the case of complex roots, let us find the zero-input response of an LTID system described by the equation

$$(E^2 - 1.56E + 0.81)\ y[k] = (E + 3)f[k]$$

When the initial conditions are $y_0[-1]$ and $y_0[-2] = 1$.

Solution :

The characteristic polynomial is $(\gamma^2 - 1.56\gamma + 0.81) = (\gamma - 0.78 - j0.45)\ (\gamma - 0.78 + j0.45)$.

The characteristic roots are $0.78 \pm j0.45$; that is, $0.9e^{\pm j\frac{\pi}{6}}$. Thus, $|\gamma| = 0.9$ and $\beta = \pi / 6$. and the zero-input response, refer above table point (3).

$$y_0[k] = c(0.9)^k \cos\left(\frac{\pi}{6}k + \theta\right)$$

To determine the arbitrary constants c and θ, we set k = –1 and –2 in this equation and substituting the initial conditions $y_0[-1] = 2, y_0[-2] = 1$, we obtain

$$2 = \frac{c}{0.9}\cos\left(-\frac{\pi}{6} + \theta\right) = \frac{c}{0.9}\left[\frac{\sqrt{3}}{2}\cos\theta + \frac{1}{2}\sin\theta\right]$$

$$1 = \frac{c}{(0.9)^2}\cos\left(-\frac{\pi}{3} + \theta\right) = \frac{c}{0.81}\left[\frac{1}{2}\cos\theta + \frac{\sqrt{3}}{2}\sin\theta\right]$$

or

$$\frac{\sqrt{3}}{1.8}c\cos\theta + \frac{1}{1.8}c\sin\theta = 2$$

$$\frac{1}{1.62}c\cos\theta + \frac{\sqrt{3}}{1.62}c\sin\theta = 1$$

These are two simultaneous equation in two unknowns c cos θ and c sin θ. Solution of these equation yields

$$c\cos\theta = 2.308$$
$$c\sin\theta = -0.397$$

Dividing c sin θ by c cos θ yields

$$\tan\theta = \frac{-0.397}{2.308} = \frac{-0.172}{1}$$

$$\theta = \tan^{-1}(-0.172) = -0.17\,\text{rad}$$

substituting $\theta = -0.17$ radian in c cos θ = 2.308 yields c = 2.34, therefore

$$y_0[k] = 2.34(0.9)^k \cos\left(\frac{\pi}{6}k - 0.17\right) \qquad k \geq 0$$

Observe that here we have used radian unit for both β and θ. We also could have used the degree unit, although it is not recommended. The important consideration is to be consistent and to use the same units for both β and θ.

The Unit Impulse Response h[k]

h[k] is the system response to input $\delta[k]$, which is zero for k > 0. We know that when the input is zero, only the characteristic modes can be sustained by the system. Therefore, h[k] must be made up of characteristic modes for k > 0. At k = 0, it may have some nonzero value A_0, so that h[k] can be expressed as

$$h[k] = A_0\delta[k] + y_n[k]$$

where, $y_n[k]$ is a linear combination of the characteristic modes and $A_0 = \frac{b_0}{a_0}$

Moreover, because h[k] is causal, we must multiply $y_n[k]$ by u[k]. Therefore,

$$h[k] = \frac{b_0}{a_0}\delta[k] + y_n[k]u[k] \qquad \ldots(46)$$

The n unknown coefficients in $y_n[k]$ (on the right-hand side) can be determined from the knowledge of n values of h[k].

Example:

Determine the unit impulse response h[k] for a system specified by the equation

$$y[k] - 0.6y[k-1] - 0.16y[k-2] = 5f[k]$$

Solution:

This equation can be expressed in the advance operator form as

$$y[k+2] - 0.6y[k+1] - 0.16y[k] = 5f[k+2]$$

or $$(E^2 - 0.6E - 0.16)\,y[k] = 5E^2 f[k] \qquad \ldots(47)$$

The characteristic polynomial is

$$\gamma^2 - 0.6\gamma - 0.16 = (\gamma + 0.2)(\gamma - 0.8)$$

The characteristic modes are $(-0.2)^k$ and $(0.8)^k$. Therefore

$$y_n[k] = c_1(-0.2)^k + c_2(0.8)^k$$

Also, from equation (47), also we have $a_0 = -0.16$ and $b_0 = 0$. Therefore, according to equation (46)

$$h[k] = [c_1(-0.2)^k + c_2(0.8)^k]u[k] \qquad \ldots(48)$$

To determine c_1 and c_2, we need to find two values of h[k] which can be found out iteratively, we get h[0] = 5 and h[1] = 3. Now setting k = 0 and 1 in equation (48) and using the fact that h[0] = 5 and h[1] = 3, we obtain

$$\left.\begin{aligned} 5 &= c_1 + c_2 \\ 3 &= -0.2c_1 + 0.8c_2 \end{aligned}\right\} \Rightarrow \begin{aligned} c_1 &= 1 \\ c_2 &= 4 \end{aligned}$$

Therefore

$$h[k] = [(-0.2)^k + 4(0.8)^k]\,u[k]$$

System Response to External Input : The zero-State Response

The zero-state response y[k] is the system response to an input f[k] when the system is in zero state. Here we follow the procedure parallel to that used in the continuous-time case

Therefore

$$y[k] = \sum_{m=-\infty}^{\infty} f[m]h[k-m] \qquad \ldots(49)$$

The summation on the right–hand side is known as the *convolution sum* of f[k] and h[k], and is represented symbolically by f[k] * h[k]

$$f[k] * h[k] = \sum_{m=-\infty}^{\infty} f[m]h[k-m]$$

Properties of the Convolution Sum

The structure of the convolution sum is similar to that of the convolution integral. Moreover, the properties of the convolution sum are similar to those of the convolution integral.

1. The Commutative Property

$$f_1[k] * f_2[k] = f_2[k] * f_1[k]$$

2. The Distributive Property

$$f_1[k] * (f_2[k] + f_3[k]) = f_1[k] * f_2[k] + f_1[k] * f_3[k]$$

3. The Associative Property

$$f_1[k] * (f_2[k] * f_3[k]) = (f_1[k] * f_2[k]) * f_3[k]$$

4. The Shifting Property

If

$$f_1[k] * f_2[k] = c[k]$$

then+

$$f_1[k-m] * f_2[k-n] = c[k-m-n]$$

5. The Convolution with an Impulse

$$f[k] * \delta[k] = f[k]$$

6. The Width Property

If $f_1[k]$ and $f_2[k]$ have lengths of m and n elements respectively, then the length of c[k] is m + n – 1 elements.

Causality and Zero–State Response

In deriving equation (49), we assumed the system to be LTI. There were no other restrictions on either the input signal or the system. In practice, almost all of the input signals are causal, and a majority of the systems are also causal. These restrictions further simplify the limits of the sum in equation (49).
Therefore, equation (49) in this case reduces to

$$y[k] = \sum_{m=0}^{k} f[m]h[k-m] \qquad \ldots(50)$$

Example

Find the (zero–state) response y[k] of an LTID system described by the equaiton

$$y[k+2] - 0.6y[k+1] - 0.16y[k] = 5f[k+2] \quad \ldots(51)$$

if the input $f[k] = 4^{-k}u[k]$.
Solution:

In order to find zero state response, first we have to find impulse response h[k].

Step 1 *:* To find h[k]
The given DE for zero-input, becomes

$$y[k+2] - 0.6y[k+1] - 0.16\,y[k] = 0$$

∴ the characteristic polynomial is

$$\gamma^2 - 0.6\gamma - 0.16 = 0$$

Solving, we get

$$\gamma = 0.8, -0.2$$

$$\therefore \quad h[k] = \frac{b_0}{a_0}f[k] + [c_1(0.8)^k + c_2(-0.2)^k]u[k]$$

as $b_0 = 0$

$$\therefore \quad h[k] = [c_1(0.8)^k + c_2(-0.2)^k]u[k] \quad \ldots(52)$$

c_1 and c_2 can be found out by knowing two values of h[k] say h[0] and h[1]
now from equation (51), we get

$$h[k+2] - 0.6\,h[k+1] - 0.16\,h[k] = 5\delta[k+2]$$

putting k = –2, we get

$$h[0] = 5\delta[0] = 5$$

and putting k = –1, we get

$$h[1] - 0.6h[0] = 0$$

$$\therefore \quad h[1] = \frac{6}{10}.5 = 3$$

using values of h[0] and h[1] and from (52), we get

$$c_1 + c_2 = 5$$

$$0.8c_1 - 0.2c_2 = 3$$

$$\therefore \quad c_1 = 4,\ c_2 = 1$$

∴ $h[k] = [4(0.8)^k + (-0.2)^k]\,u[k]$ …impulse response
∴zero-state response for input $(4)^{-k}u[k]$ is given by

$$\begin{aligned} y[k] &= h[k] * f[k] = f[k] * h[k] \\ &= (4)^{-k}u[k] * \{4(0.8)^k u[k] + (-0.2)^k u[k]\} \\ &= (0.25)^k u[k] * 4(0.8)^k u[k] + (0.25)^k u[k] * (-0.2)^k u[k] \\ &= 4\sum_{m=0}^{k}(0.25)^m(0.8)^{(k-m)} + \sum_{m=0}^{k}(0.25)^m(0.2)^{(k-m)} \\ &= 4(0.8)^k\sum_{m=0}^{k}\left(\frac{0.25}{0.8}\right)^m + (-0.2)^k\sum_{m=0}^{k}\left(\frac{0.25}{-0.2}\right)^m \end{aligned}$$

$$= 4(0.8)^k\left[\frac{(0.25)^{k+1}-(0.8)^{k+1}}{(0.8)^k(0.25-0.8)}\right]+(-0.2)^k\left[\frac{(0.25)^{k+1}-(0.2)^{k+1}}{(-0.2)^k(0.25+0.2)}\right]$$

$$= -7.27(0.25)^{k+1}+7.27(0.8)^{k+1}+2.22(0.25)^{k+1}-2.22(0.2)^{k+1}$$

$$= -1.8175(4)^{-k}+5.81(0.8)^k+0.55(4)^{-k}+0.444(-0.2)^k$$

$$y[k] = [-1.26(4)^{-k}+5.81(0.8)^k+0.444(-0.2)^k]u[k]$$

Graphical Procedure for the Convolution Sum :

The steps in evaluating the convolution sum are parallel to those followed in evaluating the convolution integral. The convoluton sum of causal signals f[k] and g[k] is given by

$$c[k]=\sum_{m=0}^{k} f[m]g[k-m]$$

The convolution operation can be performed as follows:

1. Invert g[m] about the vertical axis (m = 0) to obtain g[–m] following figure (d). Following figure (e) shows both f[m] and g[–m].
2. Time shift g[–m] by k untis to obtain g[k – m]. For k > 0, the shift is to the right (delay); for k < 0, the shift is to the left (advance). Following figures ((f) and (g)) show g[k –m] for k > 0 and for k < 0, respectively.
3. Next we multiply f[m] and g[k – m] and add all the products to obtain c[k]. The procedure is repeated for each value of k over the range – ∞ to ∞.

Example

Find c[k] = f[k] * g[k]
where f[k] and g[k] are depicted in following Figures (a) and (b), respectively.

Solution:

We are given

$$f[k] = (0.8)^k \text{ and } g[k] = (0.3)^k$$

Therefore

$$f[m] = (0.8)^m \text{ and } g[k-m] = (0.3)^{k-m}$$

Following figure (f) shows the general situation for k ≥ 0. The two functions f[m] and g[k – m] overlap over the interval 0 ≤ m ≤ k. Therefore

$$c[k]=\sum_{m=0}^{k} f[m]g[k-m]$$

$$= \sum_{m=0}^{k}(0.8)^m(0.3)^{k-m}$$

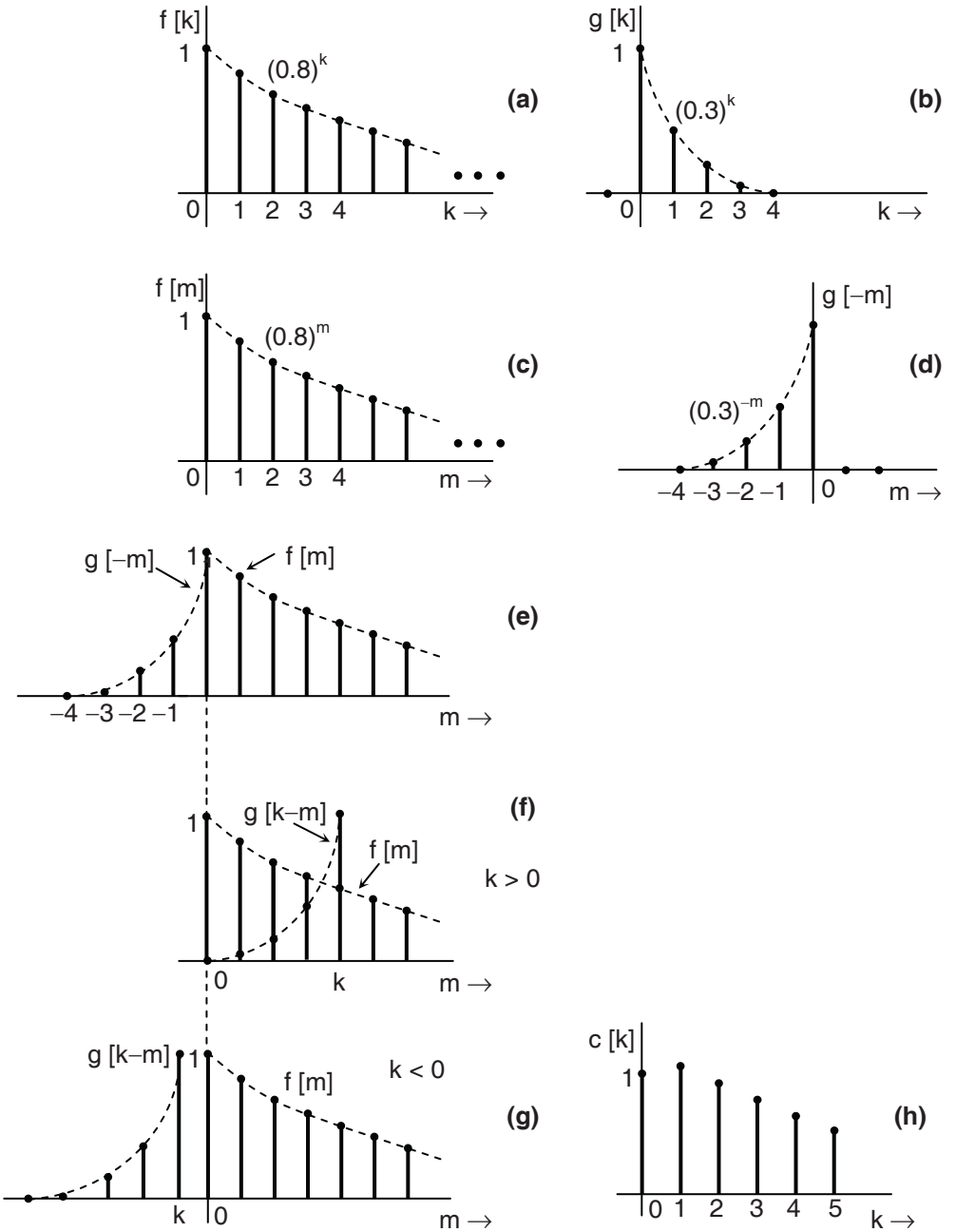
$$= (0.3)^k\sum_{m=0}^{k}\left(\frac{0.8}{0.3}\right)^m = 2\left[(0.8)^{k+1}-(0.3)^{k+1}\right]k \geq 0$$

For k < 0, there is no overlap between f[m] and g[k − m], as shown in following figure (g) so that

$$c[k] = 0 \qquad k < 0$$

and

$$\therefore \qquad c[k] = 2[(0.8)^{k+1} - (0.3)^{k+1}]\, u[k]$$



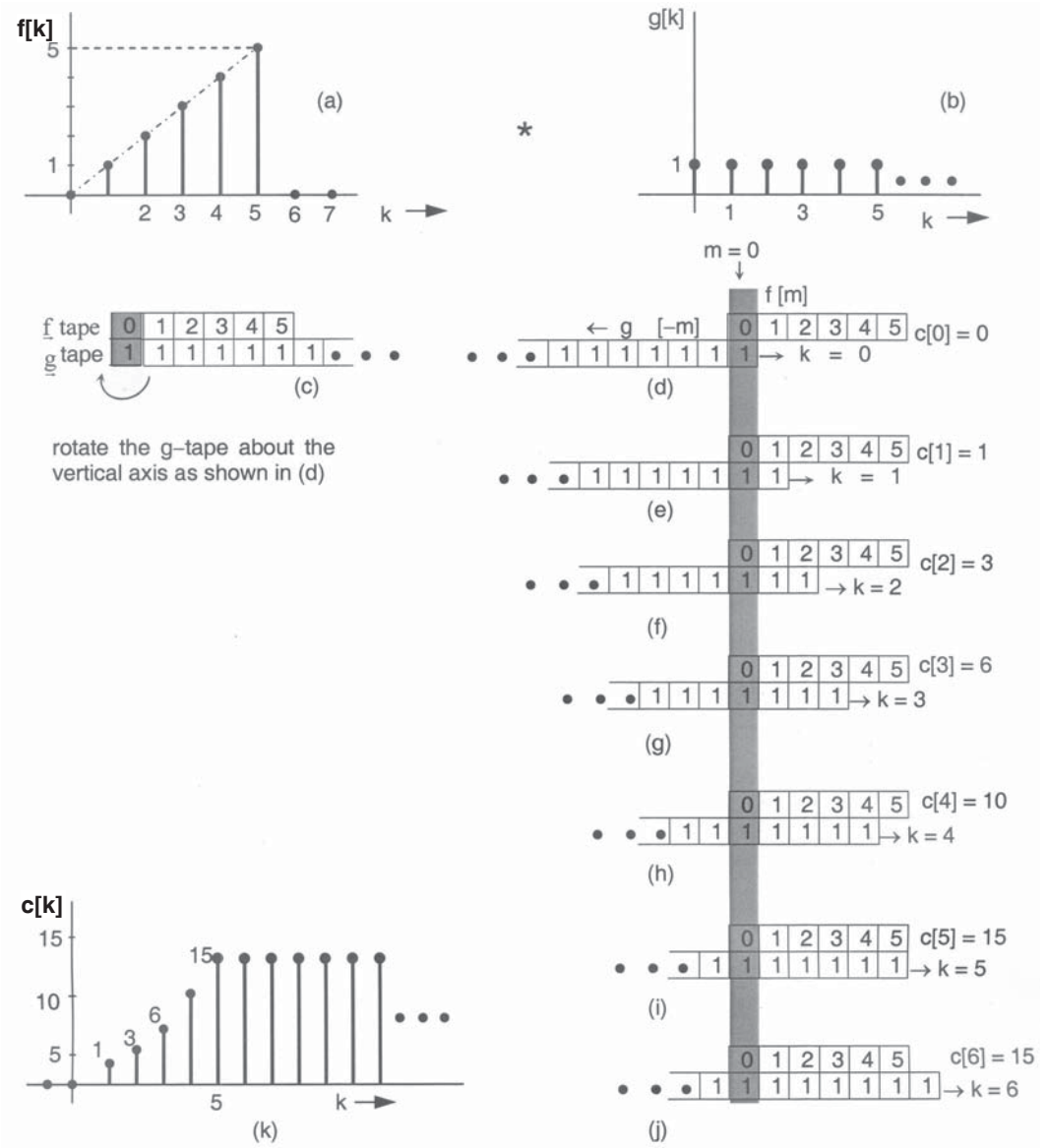
Graphical understanding of convolution of f[k] and g[k]

An Alternative Form of Graphical Procedure : The Sliding Tape Method

This algorithm is convenient when the sequences f[k] and g[k]are short or when they are available only in graphical form. This method is explained with the help of the following example

Example

Using the sliding tape method, convolve the two sequences f[k] and g[k] depicted in following
Figures (a) and (b), respectively.



Sliding tape algorithm for discrete–time convolution

Solution:

- write the sequences f[k] and g[k] in the slots of two tapes: f tape and g tape above figure (c).
- Now leave the f tape stationary (to correspond to f[m]) and invert g tape about origin, corresponding to the g[−m], above figure (d).
- Shift the inverted g tape by k slots, multiply values on two tapes in adjacent slots, and add all the products to find c[k]. Above figures (d, e, f, g, h, i and j) show the cases for k = 0, 1, 2, 3, 4, 5, and 6, respectively.

For the case of k = 0, for example, above figure (d)

$$c[0] = 0 \times 1 = 0$$

For k = 1, above figure (e)

$$c[1] = (0 \times 1) + (1 \times 1) = 1$$

Similarly,

$$c[2] = (0 \times 1) + (1 \times 1) + (2 \times 1) = 3$$
$$c[3] = (0 \times 1) + (1 \times 1) + (2 \times 1) + (3 \times 1) = 6$$
$$c[4] = (0 \times 1) + (1 \times 1) + (2 \times 1) + (3 \times 1) + (4 \times 1) = 10$$
$$c[5] = (0 \times 1) + (1 \times 1) + (2 \times 1) + (3 \times 1) + (4 \times 1) + (5 \times 1) = 15$$
$$c[6] = (0 \times 1) + (1 \times 1) + (2 \times 1) + (3 \times 1) + (4 \times 1) + (5 \times 1) = 15$$

Above figure (j) shows that c[k] = 15 for k ≥ 5. Moreover, the two tapes are nonoverlaping for k < 0, so that c[k] = 0 for k < 0. Above figure (k) shows the plot of c[k].

An Array Form of Graphical Procedure

The convolution sum can also be obtained from the array formed by sequences f[k] and g[k]. In this numerical convolution can also be perfomed from the arrays using the sets f[0], f[2],…, and g[0], g[1], g[2],…, as depicted in following figure below. The ij^{th} element (element in the ith row and j^{th} column) is given by g[i] f[j].

f[k] / g[k]	f[0]	f[1]	f[2]	f[3]	• • •
		c [0]	c [1]	c[2]	••
g [0]	f [0] g [0]	f [1] g [0]	f [2] g [0]	f [3] g [0]	• • •
g [1]	f [0] g [1]	f [1] g [1]	f [2] g [1]	f [3] g [1]	• • •
g [2]	f [0] g [2]	f [1] g [2]	f [2] g [2]	f [3] g [2]	• • •
g [3]	f [0] g [3]	f [1] g [3]	f [2] g [3]	f [3] g [3]	• • •
•	•	•	•	•	
•	•	•	•	•	
•	•	•	•	•	

Array method for computing convolution sum

We add the elements of the array along its diagonals to produce c[k] = f[k] * g[k]. For example, if we sum the elements corresponding to the first diagonal of the array, we obtain c[0]. Similarly, if we sum along the second diagonal, we obtain c[1], and so on. The process in shown above.

Example :

Convolution array method find convolution of given signal

$$x_1[k] = [1,\ 1,\ 1,\ \underset{\uparrow}{1},\ 1,\ 1,\ 1,]$$

$$x_2[k] = [1,\ 1,\ 1,\ \underset{\uparrow}{1},\ 1,\ 1,\ 1,]$$

Solution :

[Since signal are of finite duration, we can use sliding tape method or array method] here we will demonstrate array method

$x_2[k]$ \ $x_1[k]$	$x_1[-3]=1$	$x_1[-2]=1$	$x_1[-1]=1$	$x_1[0]=1$	$x_1[1]=1$	$x_1[2]=1$	$x_1[3]=1$
$x_2[-3]=1$	1	1	1	1	1	1	1
$x_2[-2]=1$	1	1	1	1	1	1	1
$x_2[-1]=1$	1	1	1	1	1	1	1
$x_2[0]=1$	1	1	1	1	1	1	1
$x_2[1]=1$	1	1	1	1	1	1	1
$x_2[2]=1$	1	1	1	1	1	1	1
$x_2[3]=1$	1	1	1	1	1	1	1

$$\therefore \quad y[k] = x_1[k] * x_2[k]$$
$$= [1, 2, 3, 4, 5, 6, \underset{\uparrow}{7}, 6, 5, 4, 3, 2, 1]$$

Total Response

The total response of an LTID system can be expressed as a sum of the zero input and zero–state components:

$$\text{Total response} = \underbrace{\sum_{j=1}^{n} c_j \gamma_j^k}_{\text{Zero–input component}} + \underbrace{f[k] * h[k]}_{\text{Zero–state component}}$$

Refering examples solved in section zero-input response and zero-state response, we get

$$\text{Total response} = \underbrace{0.2(-0.2)^k + 0.8(0.8)^k}_{\text{Zero–input component}} \quad \underbrace{-1.26(4)^{-k} + 0.444(-0.2)^k + 5.81(0.8)^k}_{\text{Zero–state component}} \quad \ldots(53)$$

Natural and Forced Response

The zero–input component is made up of characteristic modes exclusively, but the characteristic modes also appear in the zero–state response. When all the characteristic mode terms in the total response are lumped together, the resulting component is the *natural response*. The remaining part of the total response that is made up of noncharactericstic modes is the *forced response*.

∴ For above example

$$\text{Total response} = \underbrace{0.644(-0.2)^k + 6.61(0.8)^k}_{\text{Natural response}} \quad \underbrace{-1.26(4)^{-k}}_{\text{Forced response}} \qquad k \geq 0 \qquad \ldots(54)$$

Classical solution of Linear Difference Equations

If $y_n[k]$ and $y_\phi[k]$ denote the natural and the forced response respectively, then the total response is given by

$$\text{Total response} = \underbrace{y_n[k]}_{\text{modes}} + \underbrace{y_\phi[k]}_{\text{nonmodes}}$$

Because the total response $y_n[k] + y_\phi[k]$ is a solution of the system equaiton (42), we have

$$Q[E]\,(y_n[k] + y_\phi[k]) = P[E]f[k]$$

But since $y_n[k]$ is made up of characteristic modes,

$$Q[E]_{y_n}[k] = 0$$

∴we get $\quad Q[E]_{y_\phi}[k] = P[E]\,f[k] \qquad \ldots(55)$

The natural response is a linear combination of characteristic modes. The arbitraty constants (multipliers) are determined from suitable auxiliary conditions usually given as y[0], y[1], ..., y[n – 1].

Forced Response

We have shown that the forced response $y_\phi[k]$ satisfies the system Equation (55)

$$Q[E]_{y_\phi}[k] = P[E]\,f[k]$$

To determine the forced response, we shall use a method of undetermined coefficients, the same method used for the continuous–time system. These coefficients can be determned by substituting $y_\phi[k]$ in Eq. (55) and equating the coefficients of similar terms.

	Input f[k]	Forced Response $y_\phi[k]$
1.	r^k , $r \neq \gamma_i$ (i = 1, 2, ..., n)	cr^k
2.	r^k ,$r = \gamma_i$	ckr^k
3.	$\cos(\beta k + \theta)$	$c\cos(\beta k + \phi)$
4.	$\left(\sum_{i=0}^{m} \alpha_i k^i\right) r^k$	$\left(\sum_{i=0}^{m} c_i k^i\right) r^k$

Note: By definition, $y_\phi[k]$ cannot have any characteristic mode terms. If any terms shown in the right–hand column for the foced response should also be a characteristic mode of the system, the correct form of the forced response must be modified to $k^i y_\phi[k]$, where i is the smallest integer that will prevent $k^i y_\phi[k]$ from having a characteristic mode term. For example, when the input is r^k, the forced response in the right–hand column is of the form cr^k. But r^k happens to be a natural mode of the system, the correct form of the forced response is ckr^k (see Pair 2).

Example :

Find total response of

$$y[k + 2] - 0.6y[k + 1] - 0.16y[k] = 5f[k + 2]$$

if input $f[k] = 4^{-k} u[k]$ and initial conditions are $y[0] = 5.994$, $y[1] = 4.8442$

Solution:

[$\because$ initial conditions are given at y[0] and y[1], they are valid for total response]

The given equation expressed in advance operator 'E' form is

$$(E^2 - 0.6E - 0.16)\, y[k] = 5E^2 f[k]$$

$\therefore$ characteristic equation is

$$\gamma^2 - 0.6\gamma - 0.16 = 0$$
$$\gamma = 0.8, -0.2$$

$\therefore$ natural response is given by

$$y_n[k] = c_1(0.8)^k + c_2(-0.2)^k \qquad \ldots(56)$$

Now let forced response $y_\phi[k]$ be of the form $\beta(0.25)^k$ substituting in given DE, we get

$$\therefore \quad [\beta(0.25)^{(k+2)} - 0.6\beta(0.25)^{k+1} - 0.16\beta(0.25)^k] = 5(0.25)^{k+2}$$

$$\therefore \quad \beta(0.25)^k(0.25)^2 - 0.6\beta(0.25)^k(0.25) - 0.16\beta(0.25)^k = 5(0.25)^k(0.25)^2$$

$$\therefore \quad 0.0625\beta(0.25)^k - 0.15\beta(0.25)^k - 0.16\beta(0.25)^k = 0.3125(0.25)^k$$

$$-0.2475\beta = 0.3125$$
$$\beta = -1.26$$

$$\therefore \quad y_\phi[k] = -1.26\,(0.25)^k \qquad \ldots(57)$$

$\therefore$ Total response

$$y[k] = y_n[k] + y_\phi[k]$$
$$y[k] = c_1(0.8)^k + c_2(-0.2)^k - 1.26\,(0.25)^k \qquad \ldots(58)$$

now substituting initial conditions we get

$$c_1 + c_2 = 7.254$$

and $\quad 0.8c_1 - 0.2c_2 = 5.1592$

solving we get,

$$c_1 = 6.61,\ c_2 = 0.644$$

Substituting this in equation (58), we get

$$\therefore \qquad y[k] = \underbrace{6.61(0.8)^k + 0.644(-0.2)^k}_{\text{natural response}} \underbrace{-1.26(4)^{-k}}_{\text{forced response}} \quad \ldots \left[(0.25)^k = \left(\frac{1}{4}\right)^k = 4^{-k} \right]$$

Same as before the response we got in equation (54)

SYSTEM STABILITY

Just as in a continuous –time system, we define a discrete–time system to be *asymptotically stable* if, and only if, the zero–input response approaches zero as $k \to \infty$. If the zero–input response grows without bound as $k \to \infty$, the system in *unstable*.

To be more general, let γ be complex so that

$$\gamma = |\gamma| e^{j\beta} \qquad \text{and} \qquad \gamma^k = |\gamma|^k e^{j\beta k}$$

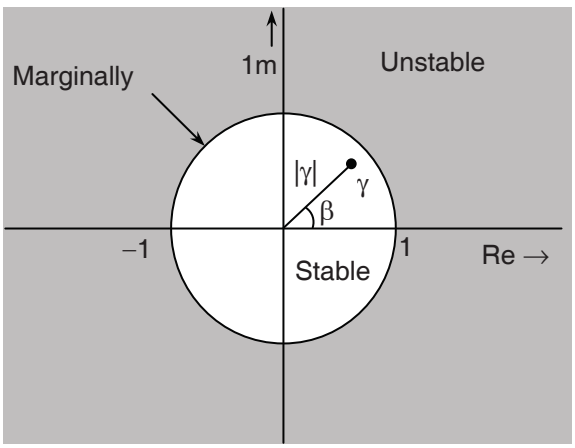
Since the magnitude of $e^{j\beta k}$ is always unity regardless of the value of k, the magnitude of γ^k is $|\gamma|^k$.

Therefore

if $|\gamma| < 1$, $\gamma^k \to 0$ as $k \to \infty$

if $|\gamma| > 1$, $\gamma^k \to \infty$ as $k \to \infty$

and if $|\gamma| = 1$, $|\gamma|^k = 1$ for all k



Characteristic roots location and system stability.

It is clear that a system is asymptotically stable if and only if

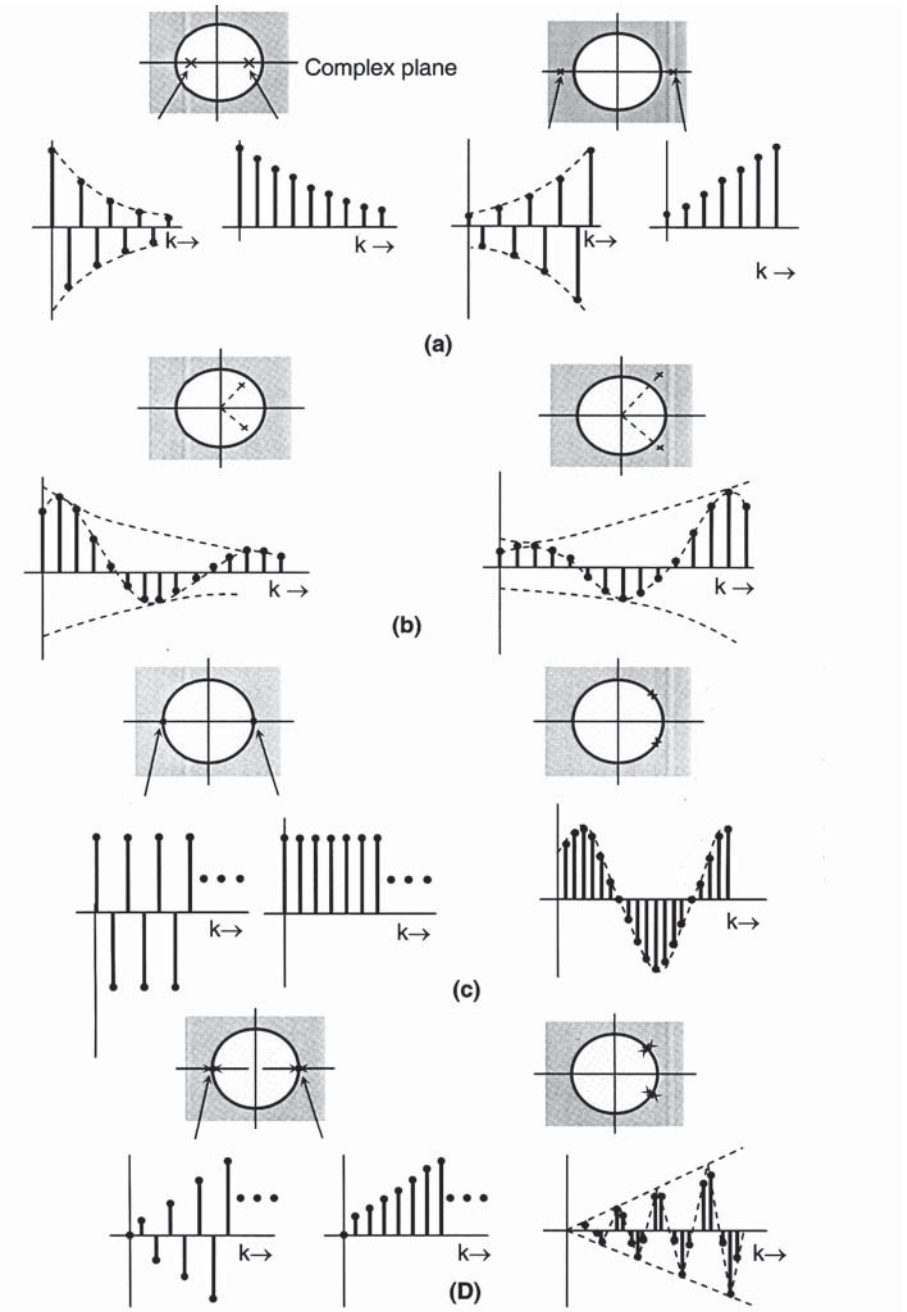
$$|\gamma_i| < 1 \qquad i = 1, 2, \ldots, n.$$

To summarize:

1. An LTID system is asymptotically stable if and only if all the characteristic roots are inside the unit circle. The roots may be simple or repeated.

2. An LTID system is unstable if and only if either one or both of the following conditions exist:
   (i) at least one root is outside the unit circle; (ii) there are repeated roots on the unit circle.

3. An LTID system is marginally stable if and only if there are no roots outside the unit circle and there are some unrepeated roots on the unit circle.



Characteristic root location and the corresponding characteristic modes

System Response to Bounded Inputs

As in the case of continuous–time systems, asymptotically stable discrete–time systems have the property that every bounded input produces a bounded output if.

$$\sum_{k=-\infty}^{\infty} |h[k]| < K_2 < \infty \qquad \ldots(59)$$

A system is said to be stable in the sense of having bounded output for every bounded input
(BIBO stability) if and only if its impulse response h[k] satisfied equation. (59).

LIST OF FORMULAE

- **Signal energy**

  For Continuous signal :

  $$E = \int_{-\infty}^{\infty} |f(t)|^2 \, dt$$

  For Discrete signal :

  $$E = \sum_{k=-\infty}^{\infty} |f[k]|^2$$

- **Signal Power**

  For Continuous signal :

  $$P_{avg} = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} |f(t)|^2 \, dt$$

  For Discrete signal :

  $$P_{avg} = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{k=-N}^{N} |f[k]|^2$$

- **Convolution :**

  For Continuous signal :

  $$y(t) = f_1(t) * f_2(t) = \int_{-\infty}^{\infty} f_1(\lambda) f_2(t-\lambda) d\lambda = \int_{-\infty}^{\infty} f_1(t-\lambda) f_2(\lambda) d\lambda$$

  i) $f(t) * \delta(t) = f(t)$

  ii) $f(t) * \delta(t - t_0) = f(t - t_0)$

  iii) $f(t - t_1) * \delta(t - t_2) = f(t - t_1 - t_2)$

  iv) $\delta(t - t_1) * \delta(t - t_2) = \delta(t - t_1 - t_2)$

For Discrete signal :

$$y[k] = f[k] * h[k] = \sum_{m=-\infty}^{\infty} f[m]h[k-m]$$

(i) $f_1[k] * f_2[k] = f_2[k] * f_1[k]$

(ii) $f_1[k] * (f_2[k] + f_3[k]) = f_1[k] * f_2[k] + f_1[k] * f_3[k]$

(iii) $f_1[k] * (f_2[k] * f_3[k]) = (f_1[k] * f_2[k]) * f_3[k]$

(iv) $f_1[k-m] * f_2[k-n] = c[k-m-n]$

(v) $f[k] * \delta[k] = f[k]$

- **Miscellaneous :**

(i) $S_k = \sum_{m=0}^{k-1} a^m = 1 + a + a^2 + \ldots + a^{k-1}$

(ii) $S_k = \dfrac{1-a^k}{1-a}$ [For G.P. with common ratio as 'a' such that | a | < 1]

(iii) $\sum_{m=0}^{k-1} a^m = k, \quad a = 1$

$= \dfrac{a^k - 1}{a-1}, \quad a \neq 1$

(iv) if |a| < 1, $S_k = \dfrac{1-a^k}{1-a}$

- **Shifting Property of impulse function**

i) $\int_{-\infty}^{\infty} f(t)\delta(t) = f(0)$

ii) $\int_{-\infty}^{\infty} f(t)\delta(t-a) = f(a)$

iii) $f(t)\,\delta(t) = f(0)\,\delta(t)$

iv) $f(t)\,\delta(t - t_0) = f(t_0)\,\delta(t - t_0)$

- **Any signal can be expressed in its even and odd components**

Even component: $f_e(t) = \dfrac{1}{2}\left(f(t) + f(-t)\right)$

Odd component: $f_o(t) = \dfrac{1}{2}\left(f(t) - f(-t)\right)$

LMR (LAST MINUTE REVISION)

- Signal cannot be energy signal and power signal simultaneously.
- All periodic signals are power signal, but converse is not true.
- Derivative of unit ramp $\rightarrow$ unit step
  Derivative of unit step $\rightarrow$ unit impulse
  Derivative of unit impulse $\rightarrow$ unit doublet
- unit step u(t) = $\int_{-\infty}^{t} \delta(t)\,dt$
- A causal signal are one which are zero for time less than zero
  i.e. for continuous f(t) = 0 , for t < 0
  for discrete f[k] = 0 , for k < 0
- All memoryless systems are causal.
- System with memory tend to be unstable.
- An initially relaxed system is a causal system.
- For linear systems, zero input yields zero output.
- A memoryless system depends for its output only at that time when the input is present. If the output is the same as the input it is called identity system.
- The discrete time unit step is the running sum of the unit sample or impulse

  $$u[k] = \sum_{m=-\infty}^{k} \delta[k]$$

  The running sum is zero for k < 0 and 1 for k $\geq$ 0 from the definition of unit impulse. The discrete time unit step can be written in terms of unit sample as

  $$u[k] = \sum_{m=0}^{\infty} \delta[k-m]$$

- Some useful parties of convolution integral are :
  (i) Commutative (ii) Distributive (iii) associative (iv) width property
- Non–periodic functions such as unit ramp and unit step are called singularity or generalized functions.
- If impulse response is not absolutely integrable, the system is unstable.
- For system to be absolutely stable all its characteristic modes must $\rightarrow$ 0 as t $\rightarrow \infty$
  i.e. (i) for LTIC system, all characteristic roots must lie in the LH-side of S-plane
  (ii) for LTID system, all characteristic roots must lie in side unit circle of Z-plane
- Classical solution of differential/difference equation given total response in terms of natural and forced response.
- Homogeneous solution is often referred to as natural response and forced response is also called particular integral.
- A system is said to be relaxed if the energy storing elements alias memory elements are set to zero before excitation.
- Linearity indicates additivity, scaling homogeneity (i.e. satisfies superposition theorem).

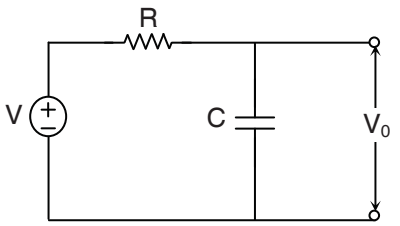
❑ ❑ ❑ ❑ ❑ ❑

ASSIGNMENT – 1

Duration : 45 mins **Marks : 30**

Q1 to Q6 carry one mark each

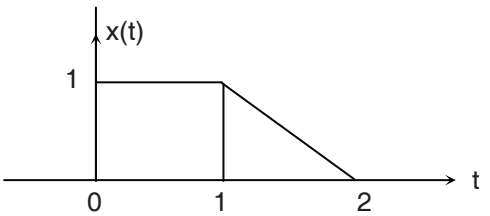
1. The following circuit is :



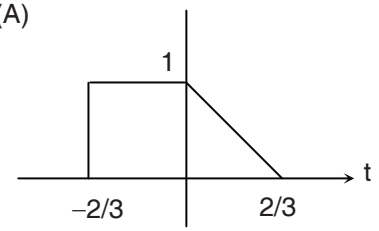
(A) Memory less system (B) Causal system
(C) Static system (D) Dynamic system

2. For a continuous time system, to be BIBO stable _____.
(A) the impulse response of the system should be absolutely integrable
(B) the impulse response of the system should be absolutely summable
(C) the o/p of the system depends upon present and past inputs
(D) the o/p of the system depends upon present, past and future inputs

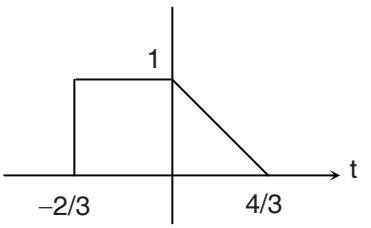
3. Find the signal $x\left(1-\frac{3}{2}t\right)$



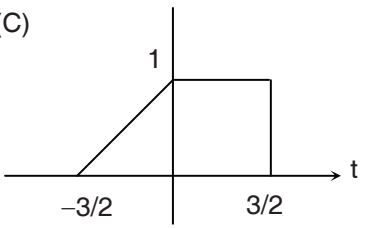
(A)



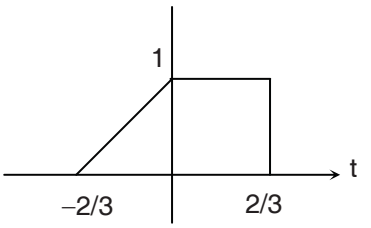
(B)



(C)



(D)



4. A periodic function of half wave symmetry is necessarily _____.

(A) an even function (B) an odd function

(C) neither odd nor even (D) both odd and even

5. About the fourier series expansion of a periodic function it can be said that

(A) Even functions have only a constant and cosine terms in their FS expansion.

(B) Odd functions have only sine terms in their FS expansion.

(C) Functions with half wave symmetry contain only odd harmonics.

(D) All of above

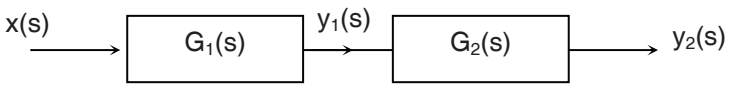
6. A periodic signal has power P/4 equal to average energy per period then RMS value of signal is given by _____

(A) $\frac{\sqrt{P}}{2}$ (B) P/4

(C) P/2 (D) $\frac{\sqrt{P}}{4}$

Q7 to Q18 carry two marks each

7.



For the above system

$$G_1(s) = \frac{s^{-1}}{1+2s^{-1}+s^{-2}}$$

Find $G_2(s)$ for the given system to be invertible.

(A) $\frac{s}{1+2s+s^2}$ (B) $\frac{1}{1+2s+s^2}$

(C) $\frac{1+2s+s^2}{s}$ (D) $1+2s+s^2$

8. If $x[k] = \cos\left[\frac{4\pi k}{21}\right]$, this sequence will be periodic after N_0 samples. The value of N_0 is

(A) 12 (B) 24

(C) 21 (D) 2

9. Consider a discrete time signal x[k] defined by

$$x[k] = \begin{cases} 1 & -2 \le k \le 2 \\ 0 & |k| > 2 \end{cases}$$ then y[k] = x[3k – 2] is :

(A) $y[k] = \begin{cases} 1 & k = 0, 1 \\ -1 & \text{otherwise} \end{cases}$

(B) $y[k] = \begin{cases} 1 & k = 1 \\ -1 & k = -1 \end{cases}$

(C) $y[k] = \begin{cases} 1 & k = 0, 1 \\ 0 & \text{otherwise} \end{cases}$

(D) none of these

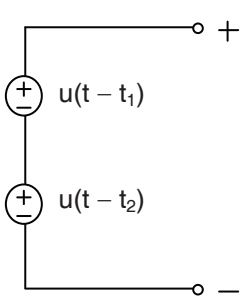
10. Match the following :

i)	Unit step function	(a)	$\pi\delta(\omega) + \frac{1}{j\omega}$
ii)	Exponential signal	(b)	$\frac{2a \sin \omega a}{\omega a}$
iii)	Signum function	(c)	$\frac{1}{a + j\omega}$
iv)	Gate function	(d)	$\frac{2}{j\omega}$

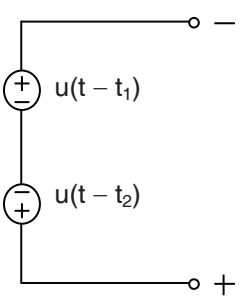
(A) (i) – (a), (ii) – (d), (iii) – (c), (iv) – (b)
(B) (i) – (d), (ii) – (c), (iii) – (a), (iv) – (b)
(C) (i) – (b), (ii) – (c), (iii) – (a), (iv) – (d)
(D) (i) – (a), (ii) – (c), (iii) – (d), (iv) – (b)

11. Gate function is represented as _____ for $t_1 < t_2$.

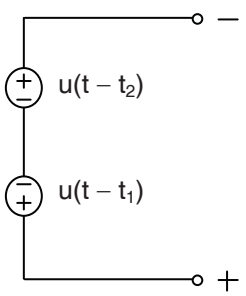
(A)



(B)

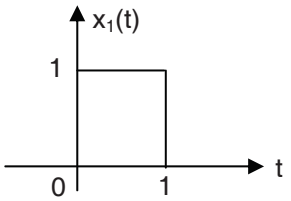


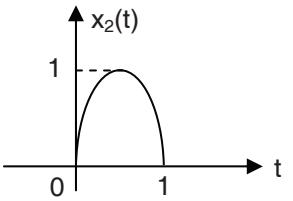
(C)



(D) None of these

12. Consider the signals as shown below





The energy in combined signal (i.e. of addition of above two signals) is given by _____

(A) 2 (B) 3.5

(C) 2.75 (D) 1

13. Consider a signal x(t) with energy $\left(\frac{E'}{2}\right)$ then time scaling by a factor 2 (i.e. doubling) will change energy to _____.

(A) E′/2 (B) E′

(C) E′/4 (D) 2E′

14. For the signal given below :

$$y(t)=2\sin\left(\frac{2}{3}t\right)+4\sin\left(\frac{1}{4}t-\frac{\pi}{4}\right)+6\sin\left(\frac{1}{3}t-\frac{\pi}{5}\right)+8\sin\left(\frac{1}{2}t-\frac{\pi}{7}\right)$$

The common period of y(t) is given by _____

(A) 12π (B) 8π

(C) 24π (D) 16π

15. The value of integral $\int_0^\infty 4t^2\delta(t+1)dt$ is

(A) 4 (B) –4

(C) 4/3 (D) None of these

16. Match the following :

	List I		**List II**
(1)	$y''(t)-2ty'(t)=x(t)$	(i)	causal and instantaneous but not linear
(2)	$y(t)=x(t)+3$	(ii)	causal and dynamic
(3)	$y(t)=2(t+1)x(t)$	(iii)	causal and instantaneous but time varying
		(iv)	causal and instantaneous but time invariant

(A) 1 – (i), 2 – (ii), 3 – (iii) (B) 1 – (ii), 2 – (i), 3 – (iii)

(C) 1 – (ii), 2 – (iii), 3 – (iv) (D) 1 – (i), 2 – (ii), 3 – (iv)

17. The signal energy in range $-1 \leq \omega \leq 1$ rad/s for $x(t) = e^{-t} u(t)$ is _____.

(A) 1/2 (B) 1/6
(C) 1 (D) ¼

18. Two statements are made regarding Response of LTI system described by difference equation

(1) The input terms of difference equation completely determine the forced response

(2) Initial conditions satisfy the total response to yield the constants in the natural response.

(3) The input terms of difference equation completely determines the natural response

Choose the correct options :

(A) 1 – True, 2 – True, 3 – False

(B) 1 – False, 2 – True, 3 – True

(C) 1 – False, 2 – True, 3 – False

(D) 1 – True, 2 – False, 3 – False

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